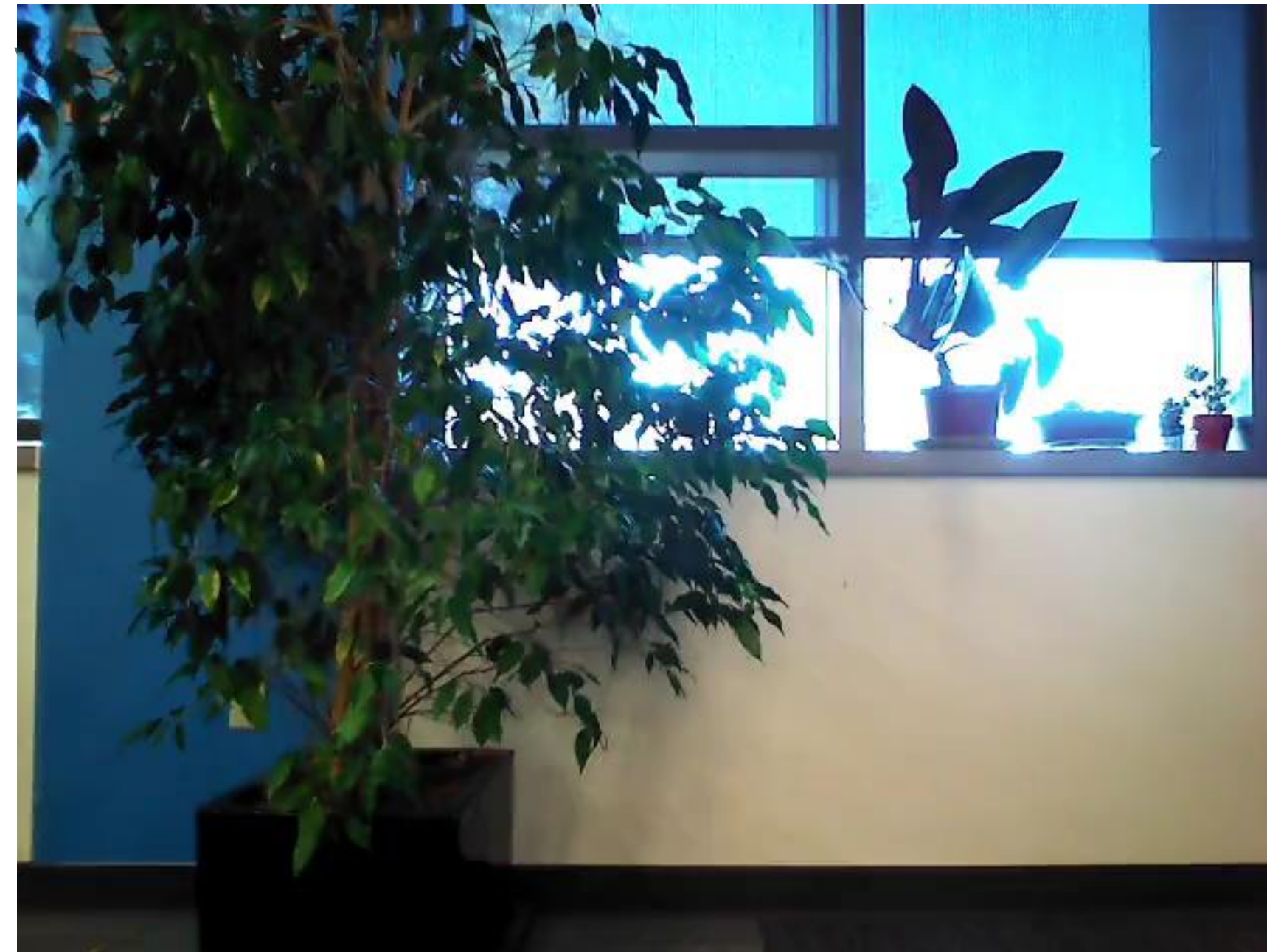
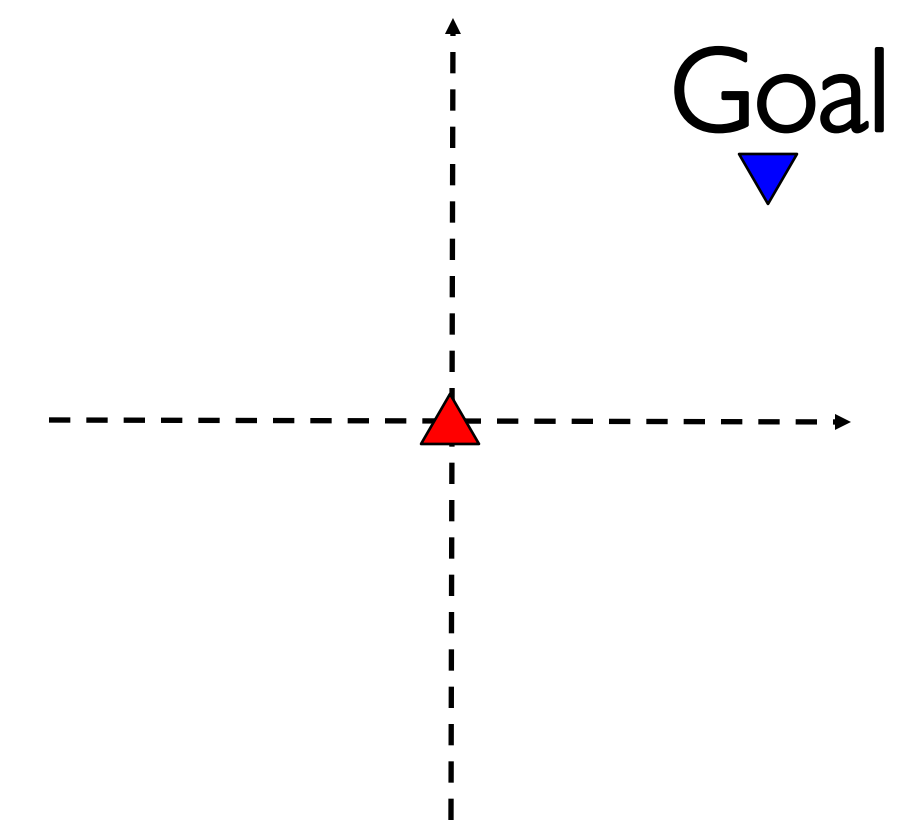




Robot with a first
person camera



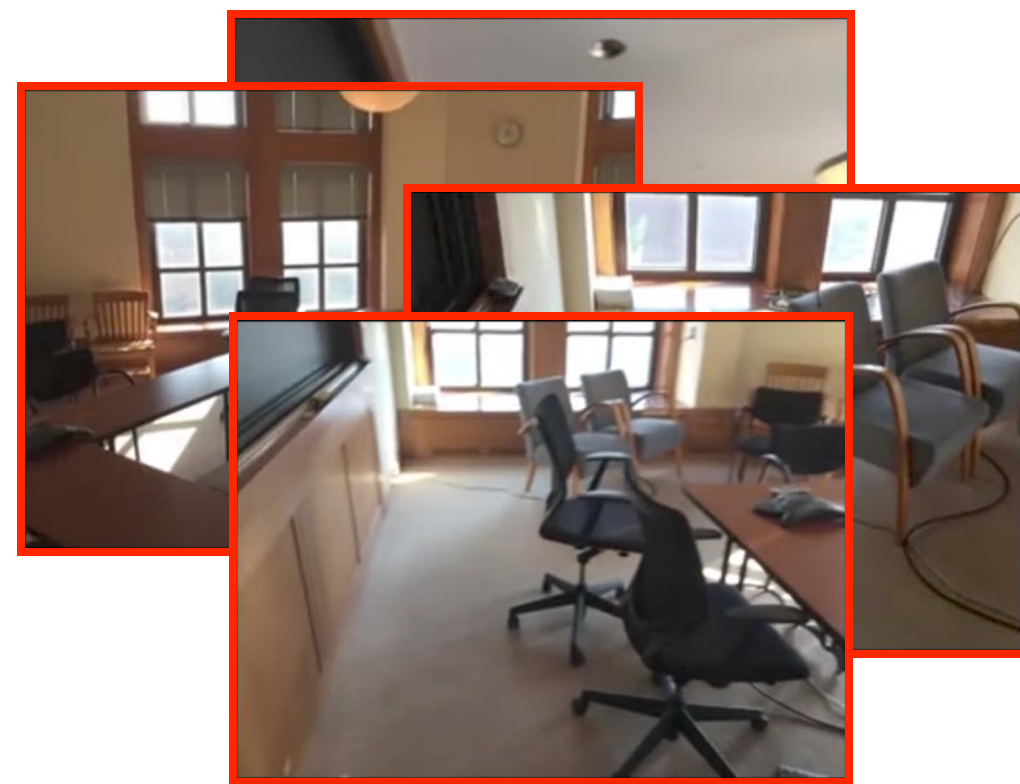
Dropped into a novel
environment



**“Go
300 feet North,
400 feet East”**

**“Go Find a
Chair”**

Navigate around



Observed Images

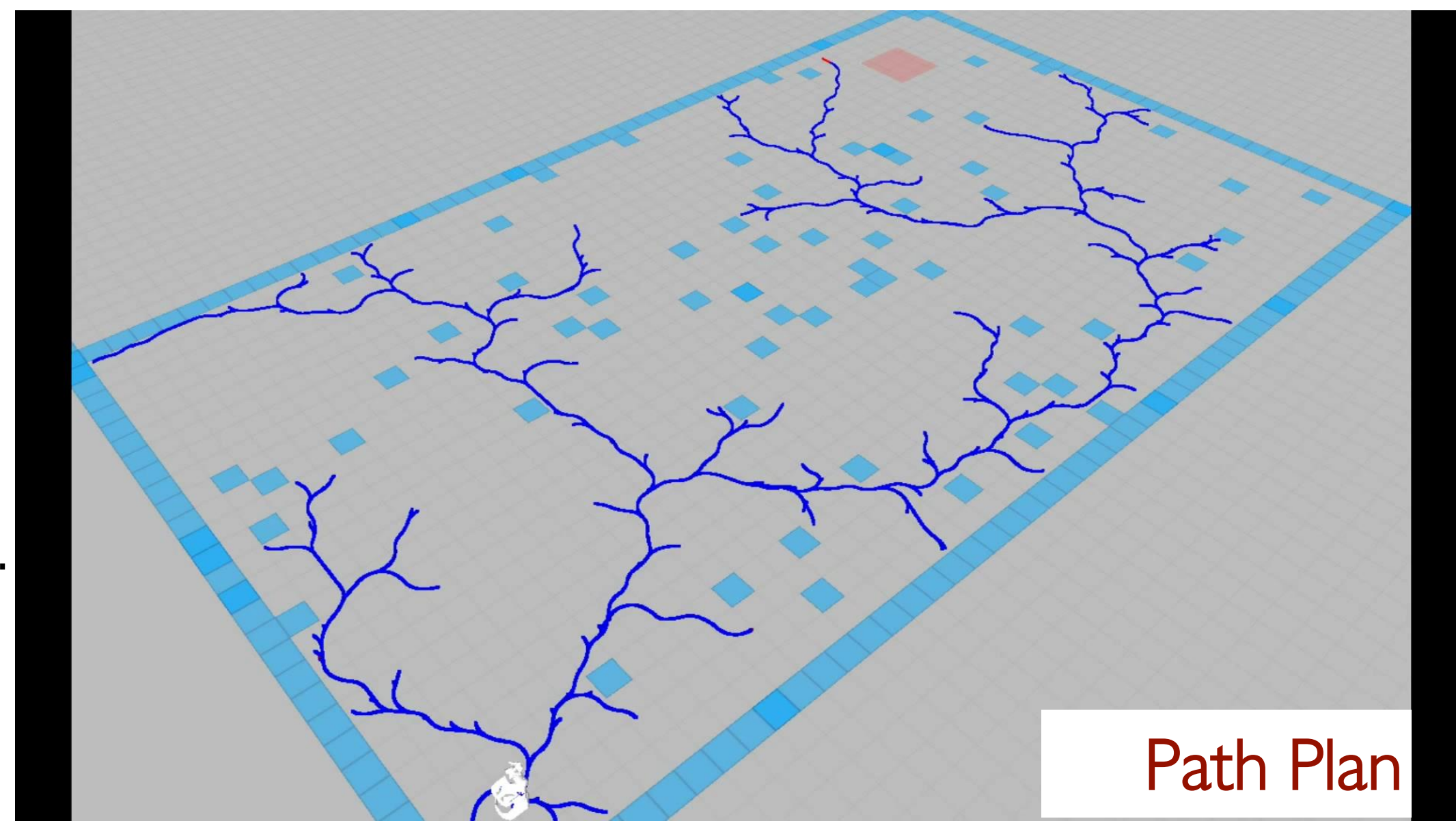
Mapping

Planning

Classical Solution



Geometric Reconstruction

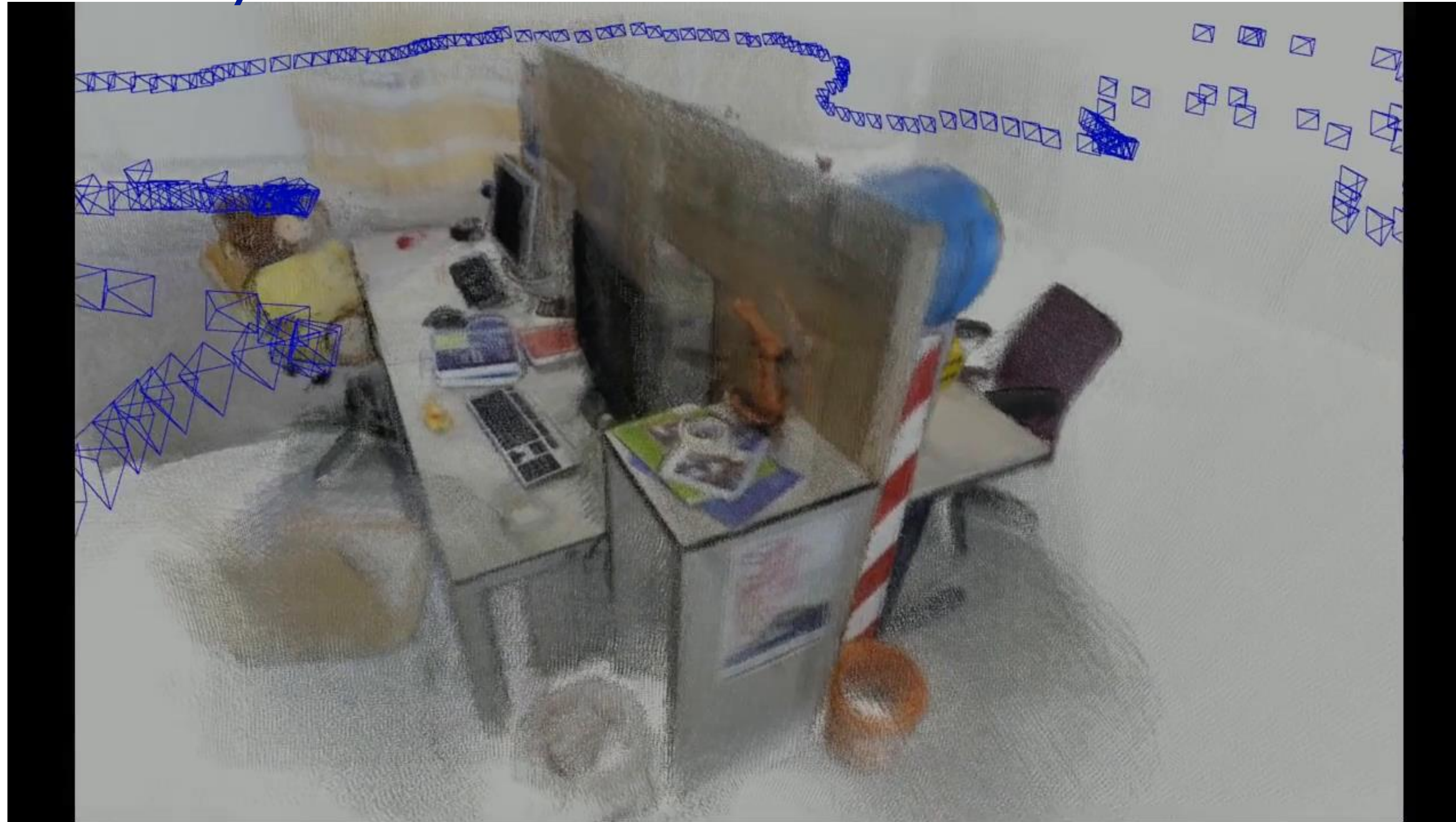


Path Plan

- Hartley and Zisserman. 2000.** Multiple View Geometry in Computer Vision
- Thrun, Burgard, Fox. 2005.** Probabilistic Robotics
- Canny. 1988.** The complexity of robot motion planning.
- Kavraki et al. RA1996.** Probabilistic roadmaps for path planning in high-dimensional configuration spaces.
- Lavalle and Kuffner. 2000.** Rapidly-exploring random trees: Progress and prospects.

Video Credits: Mur-Artal et al., Palmieri et al.

Unnecessary



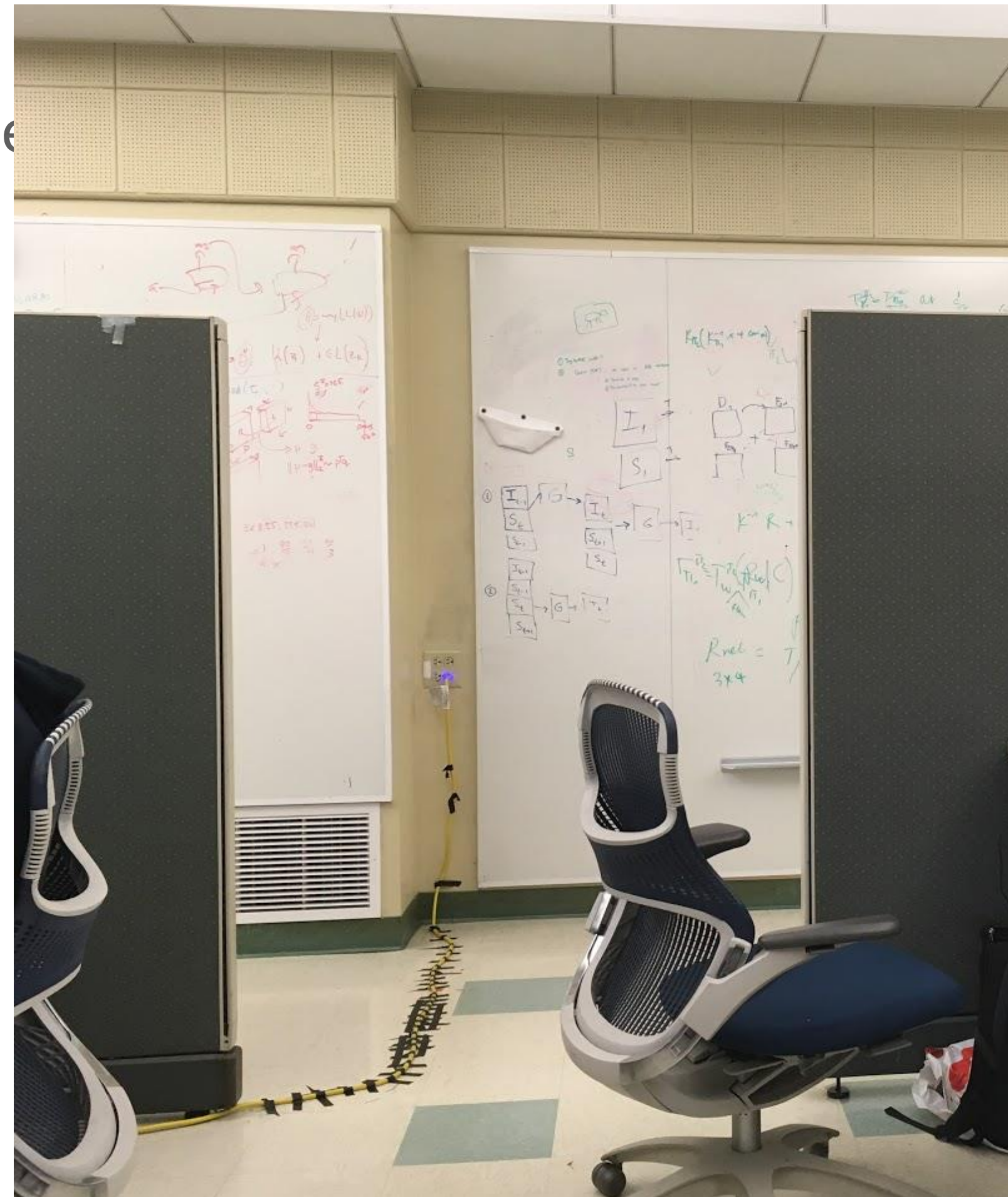
Do we need to tediously reconstruct everything on this table?

Video Credit: Mur-Artal and Tardos, TRobotics 2016. ORB-SLAM2: an Open-Source SLAM System for Monocular, Stereo and RGB-D Cameras.

Insufficient

Ge

World



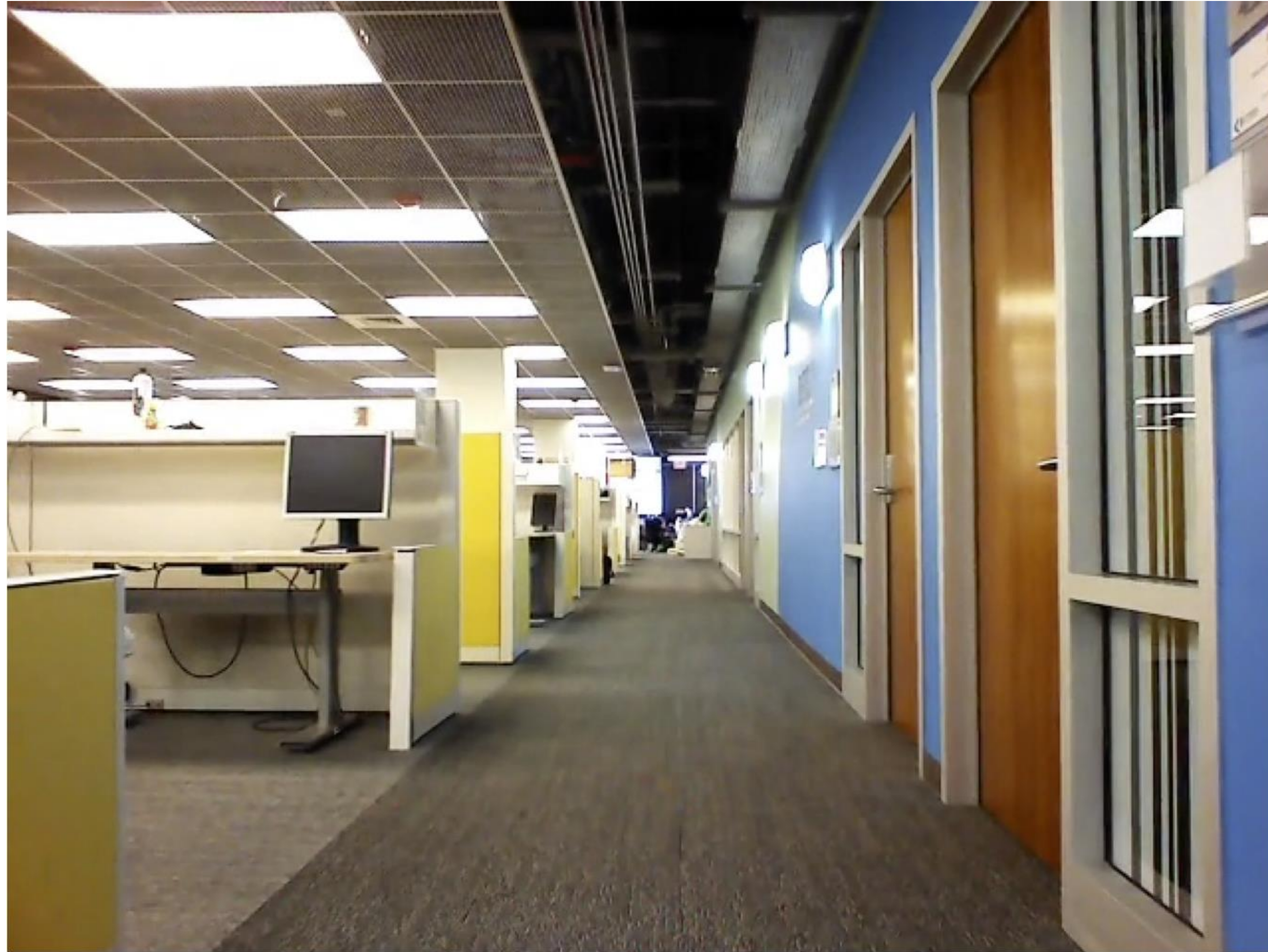
Can't speculate about space not directly observed.

Insufficient



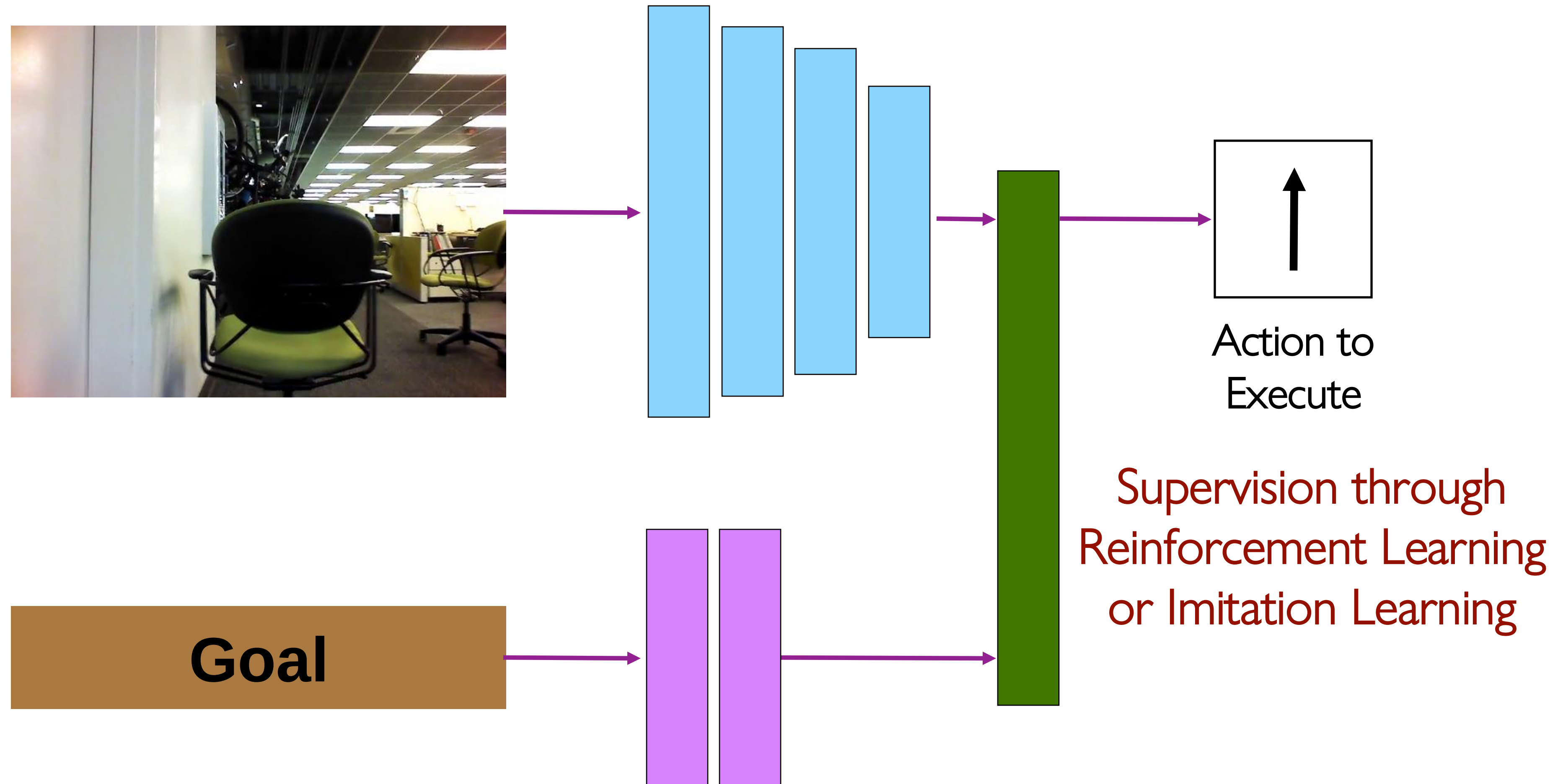
Can't exploit patterns in layout of indoor spaces.

Insufficient



Ignore navigational affordances.

Modern End-to-End Learned Navigation

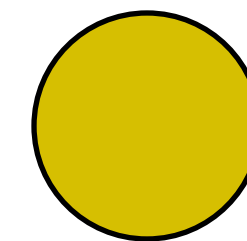
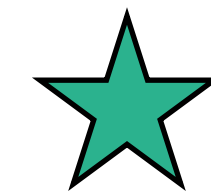
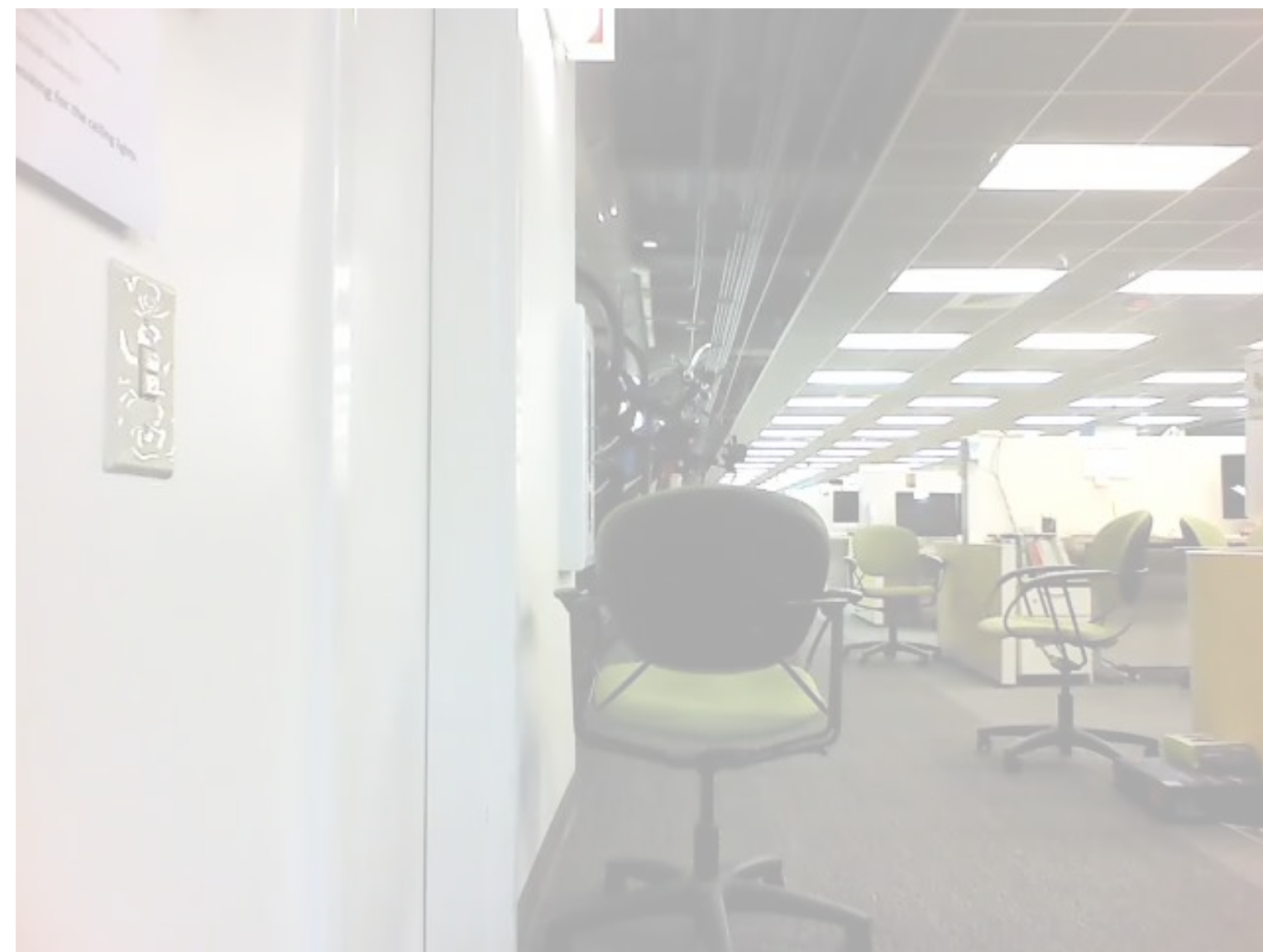


Mnih et al., Nature 2014. Human-level control through deep reinforcement learning.

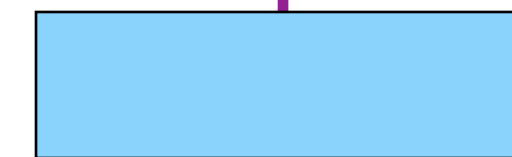
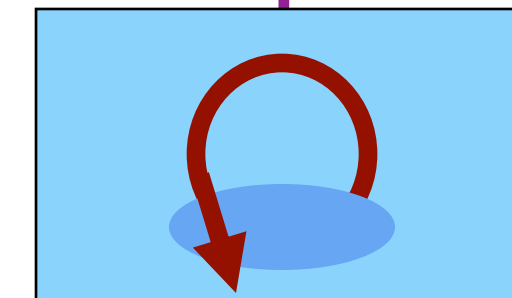
Levine et al., JMLR 2015. End-to-End Training of Deep Visuomotor Policies.

Zhu et al., ICRA 2017. Target-driven Visual Navigation in Indoor Scenes using Deep Reinforcement Learning.

Simple networks are not enough



Action to
Execute



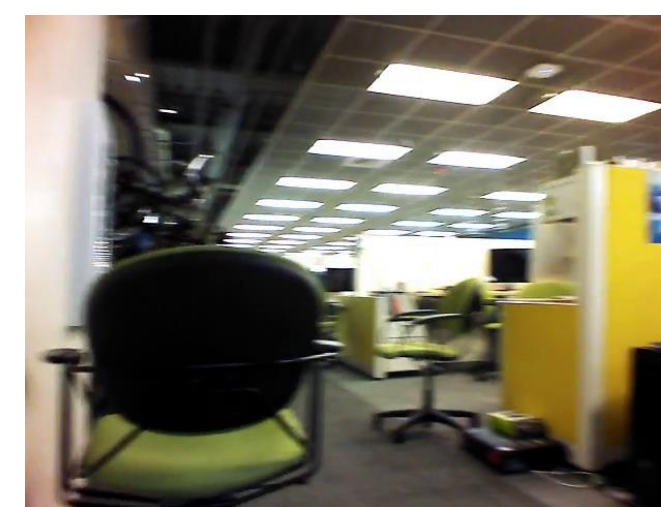
$t-4$

$t-3$

$t-2$

$t-1$

t



Mnih et al., Nature 2014. Human-level control through deep reinforcement learning.

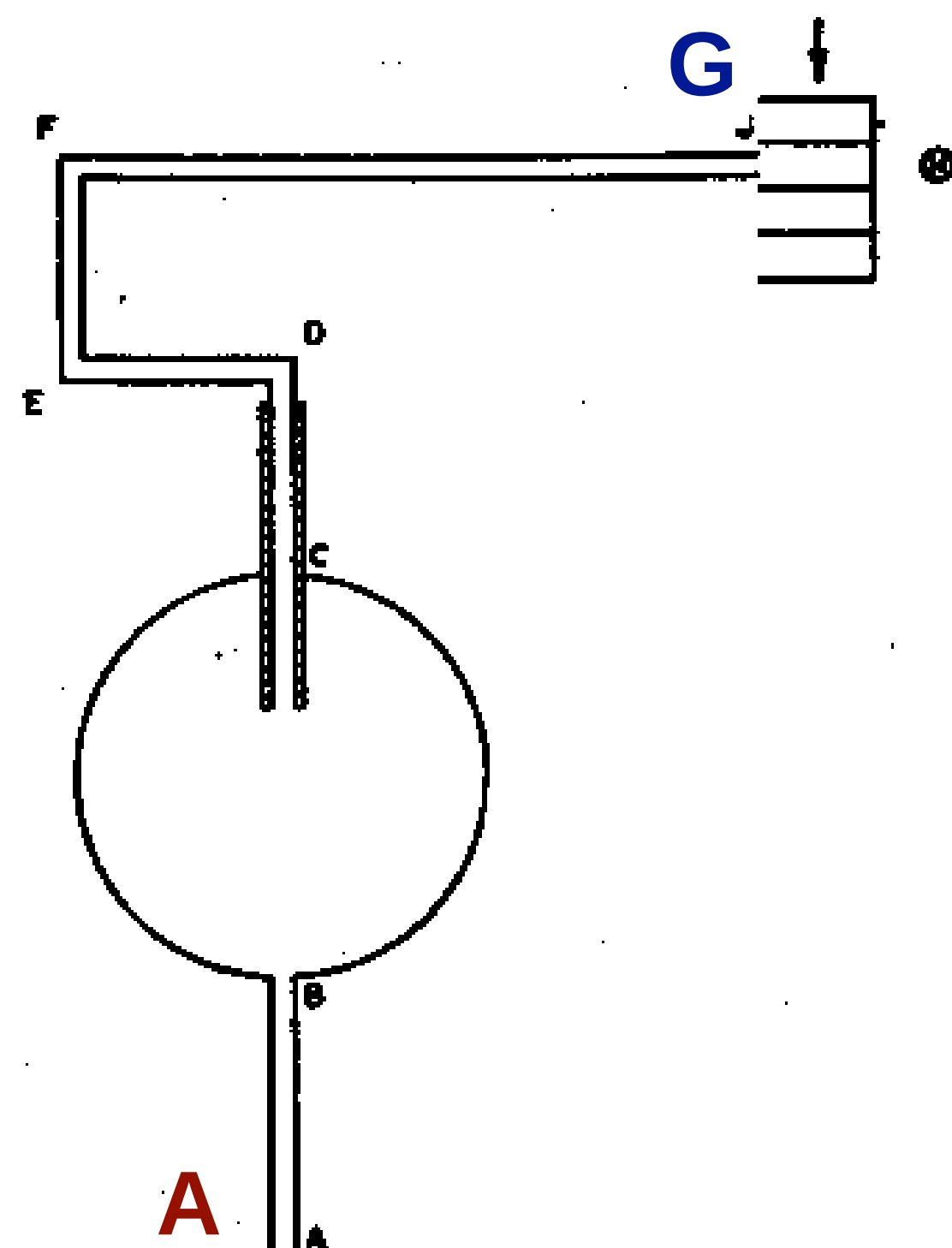
Levine et al., JMLR 2015. End-to-End Training of Deep Visuomotor Policies.

Zhu et al., ICRA 2017. Target-driven Visual Navigation in Indoor Scenes using Deep Reinforcement Learning.

COGNITIVE MAPS IN RATS AND MEN ¹

BY EDWARD C. TOLMAN

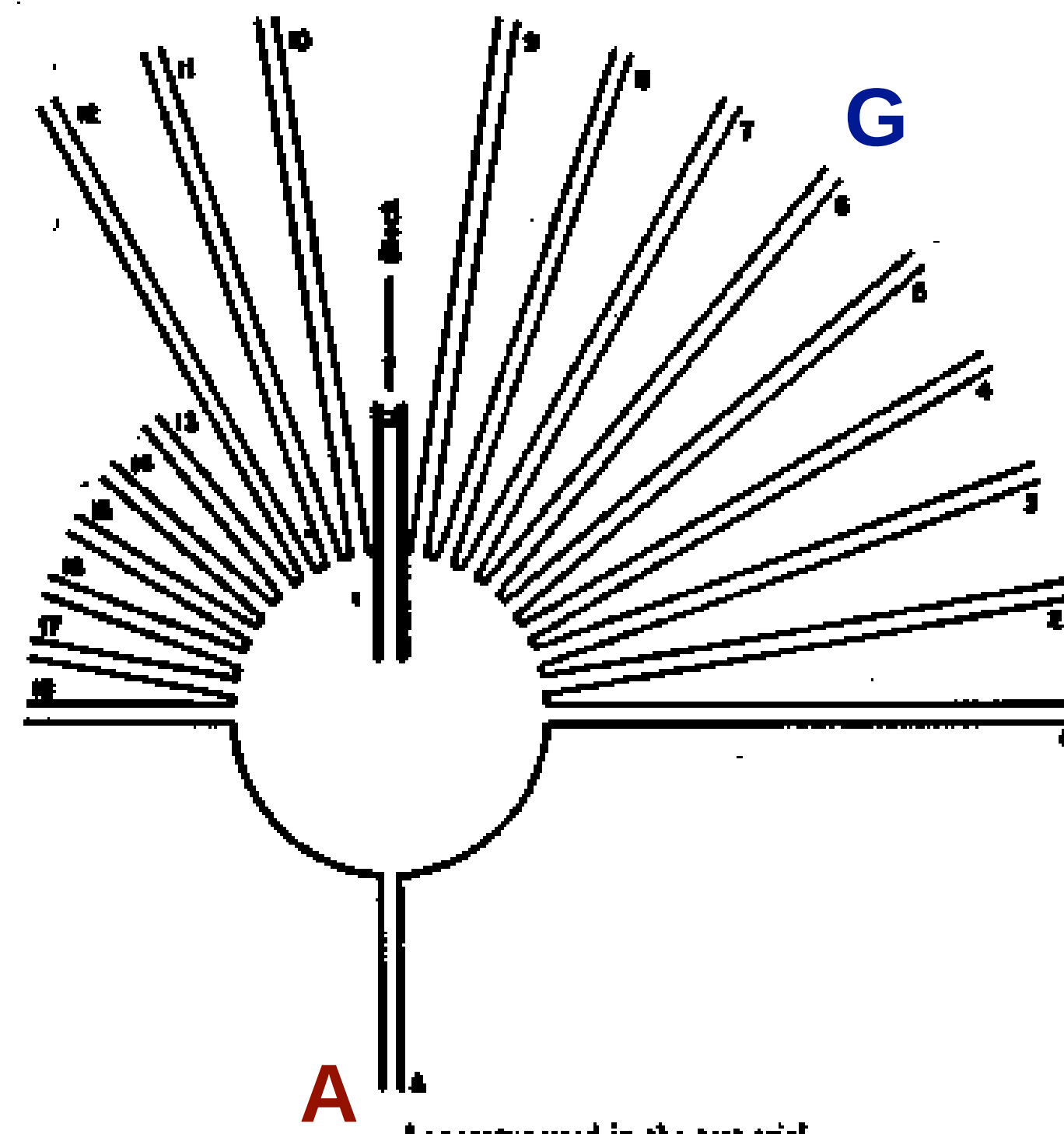
University of California



Apparatus used in preliminary training

FIG. 15

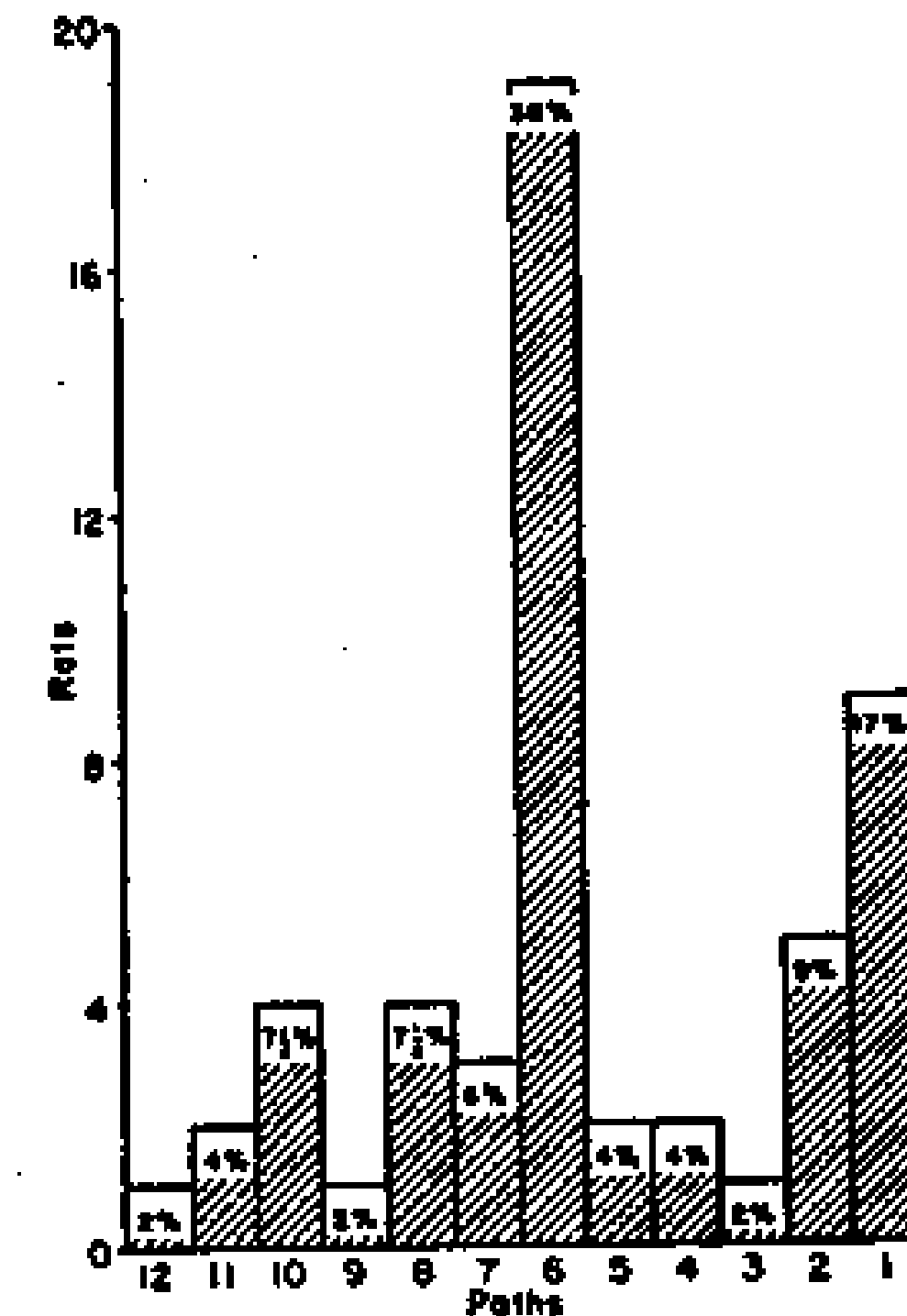
(From E. C. Tolman, B. F. Ritchie and D. Kalish, Studies in spatial learning. I. Orientation and the short-cut. *J. exp. Psychol.*, 1946, 36, p. 16.)



Apparatus used in the test trial

FIG. 16

(From E. C. Tolman, B. F. Ritchie and D. Kalish, Studies in spatial learning. I. Orientation and short-cut. *J. exp. Psychol.*, 1946, 36, p. 17.)



Numbers of rats which chose each of the paths

FIG. 17

(From E. C. Tolman, B. F. Ritchie and D. Kalish, Studies in spatial learning. I. Orientation and the short-cut. *J. exp. Psychol.*, 1946, 36, p. 19.)

Navigating to Objects in the Real World



Theophile Gervet ¹



Soumith Chintala ⁴



Dhruv Batra ^{3,4}



Jitendra Malik ^{2,4}



Devendra Chaplot ⁴

¹ **Carnegie
Mellon
University**

²  **Berkeley**
UNIVERSITY OF CALIFORNIA

³ **Georgia
Tech** 

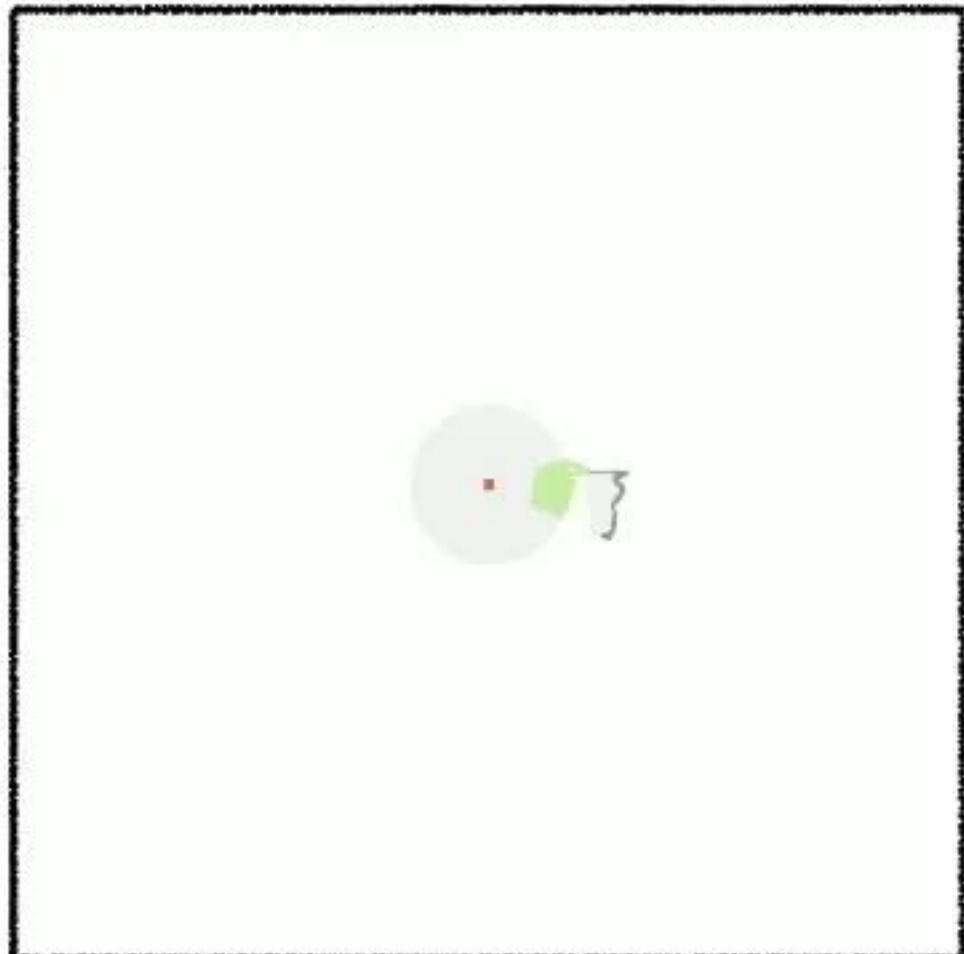
⁴  **Meta AI**

Observation



Goal: *plant*

Predicted Semantic Map



Third-person view



0: Navigable Area

1: chair

2: couch

3: potted plant

4: bed

5: toilet

6: tv

7: dining-table

8: oven

9: sink

10: refrigerator

11: book

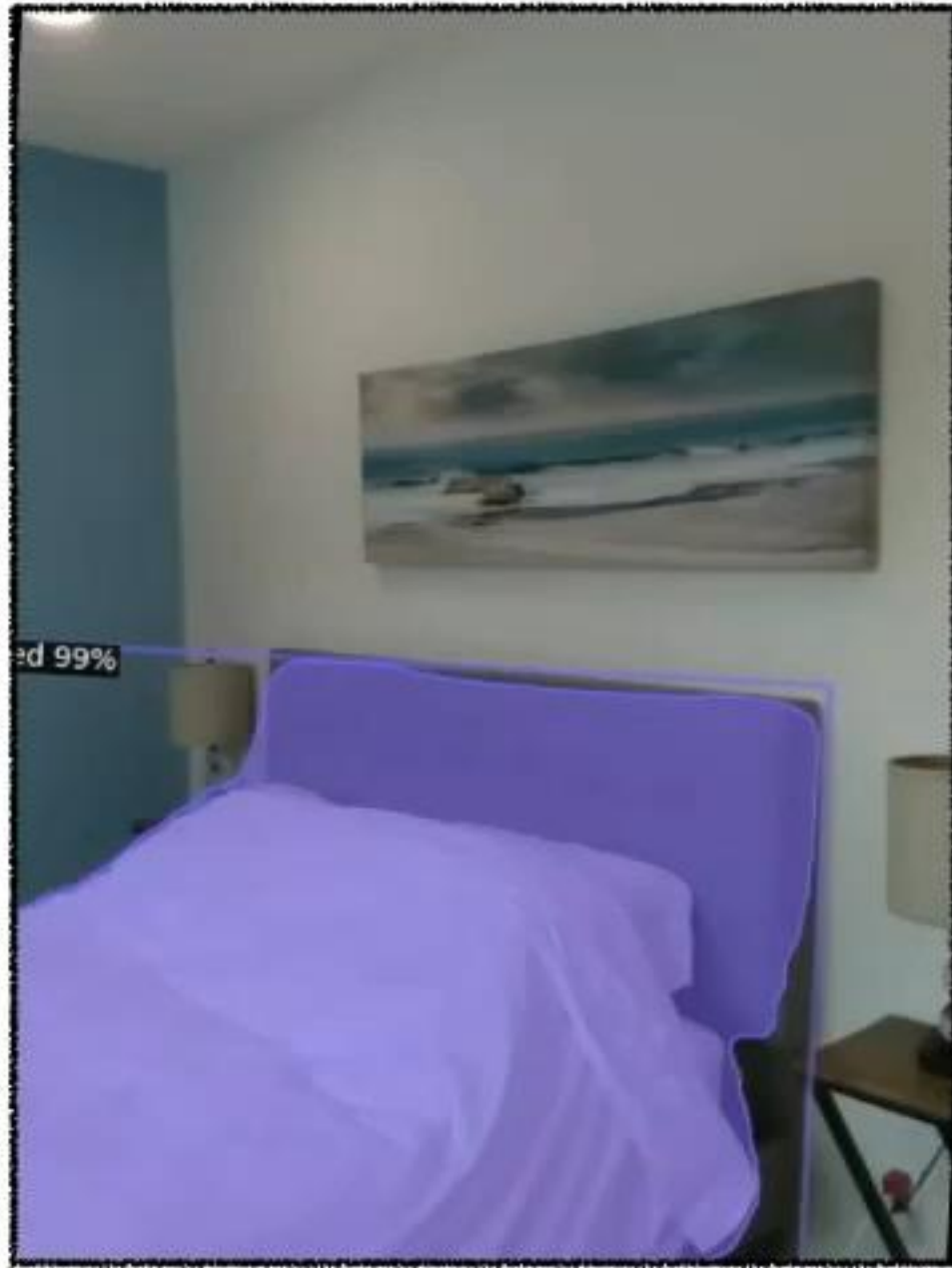
12: clock

13: vase

14: cup

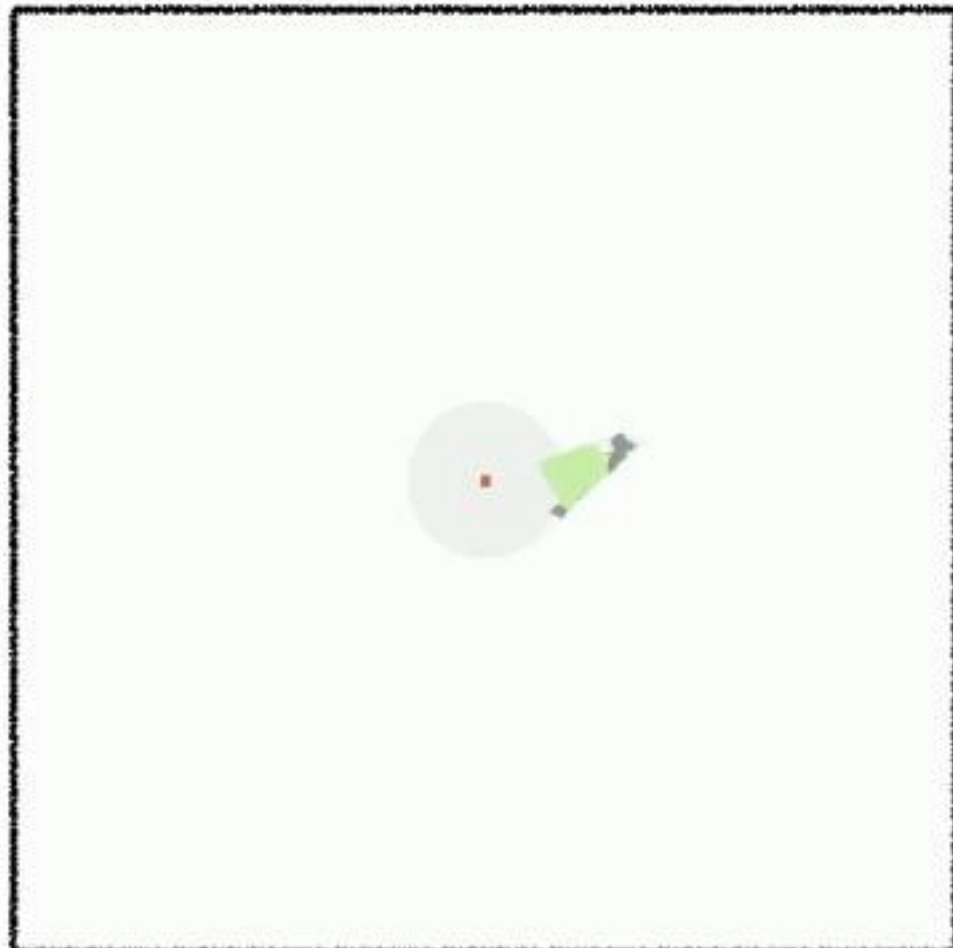
15: bottle

Observation



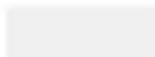
Goal: *chair*

Predicted Semantic Map



Third-person view



 Navigable Area	 3: bed	 7: oven	 11: clock
 0: chair	 4: toilet	 8: sink	 12: vase
 1: couch	 5: tv	 9: refrigerator	 13: cup
 2: potted plant	 6: dining-table	 10: book	 14: bottle

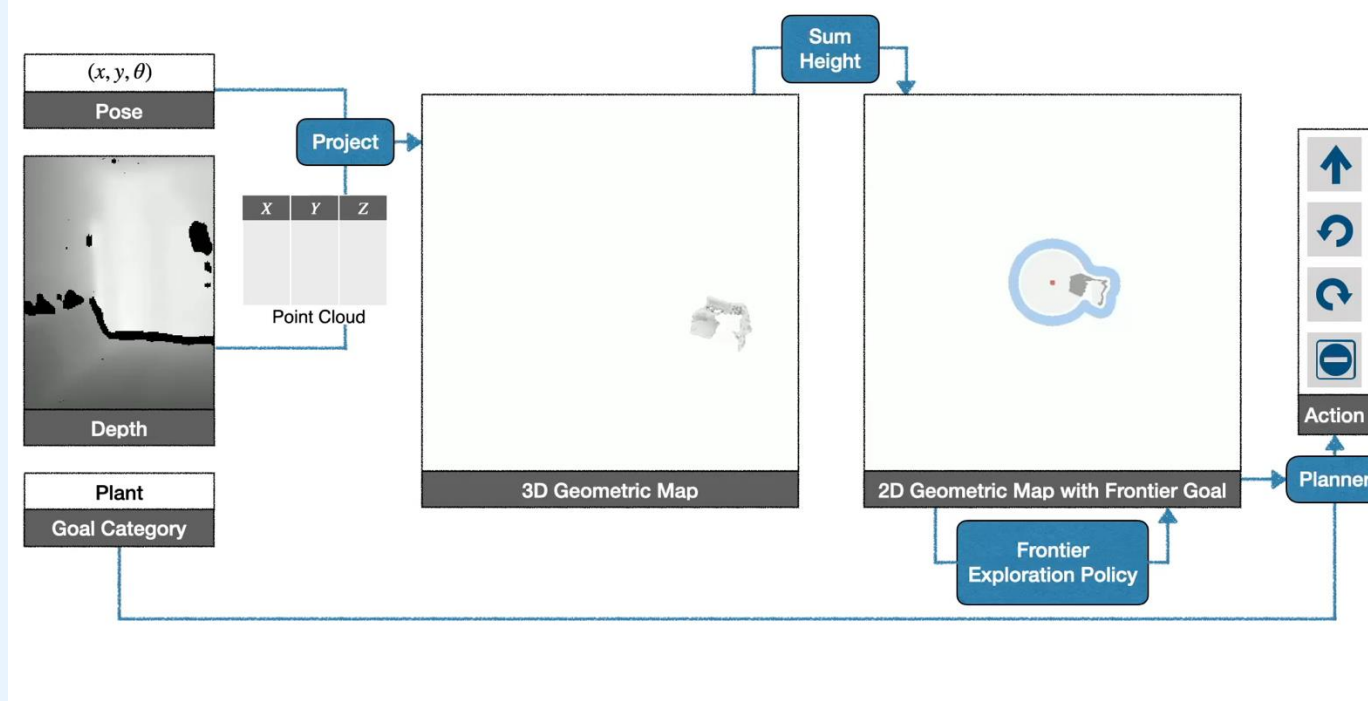


Empirical Evaluation
3 Approaches
6 Unseen Homes
6 Goal Object Categories



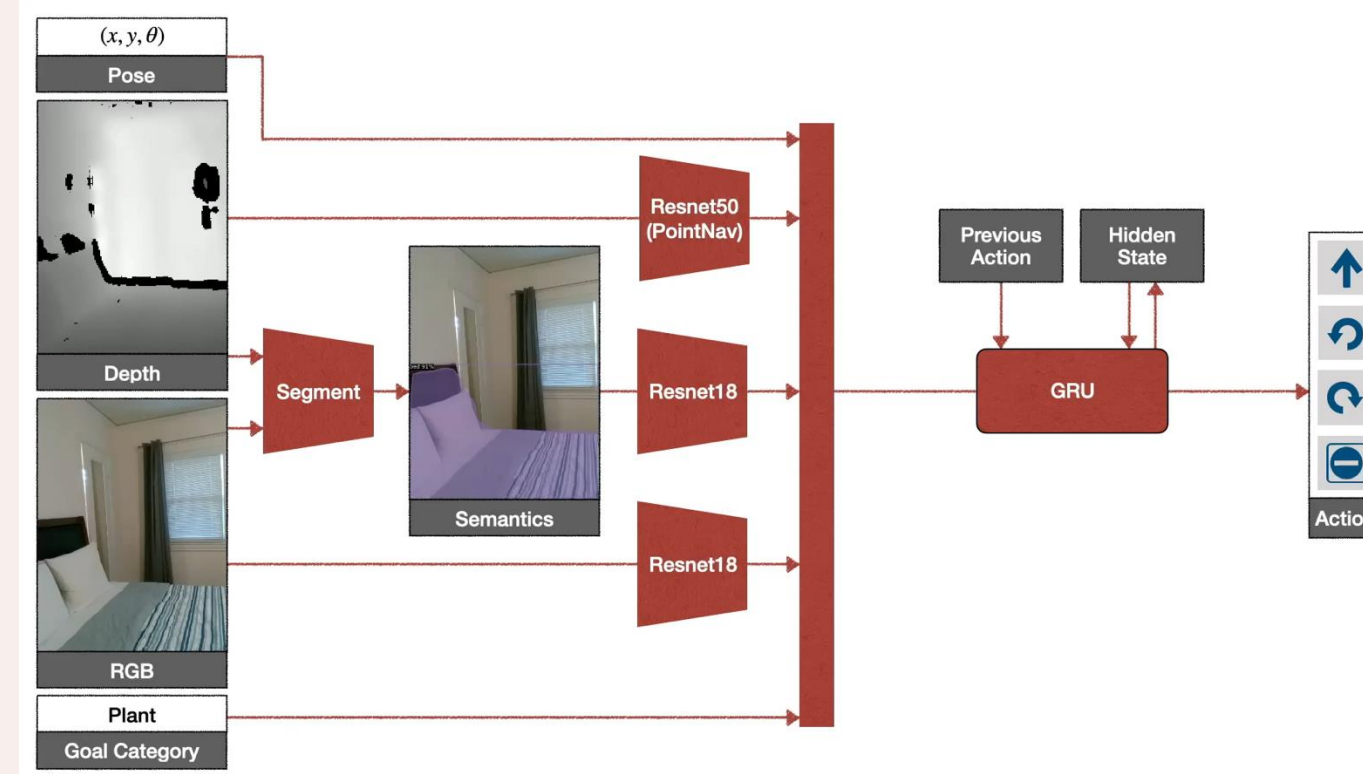
Methods

Classical



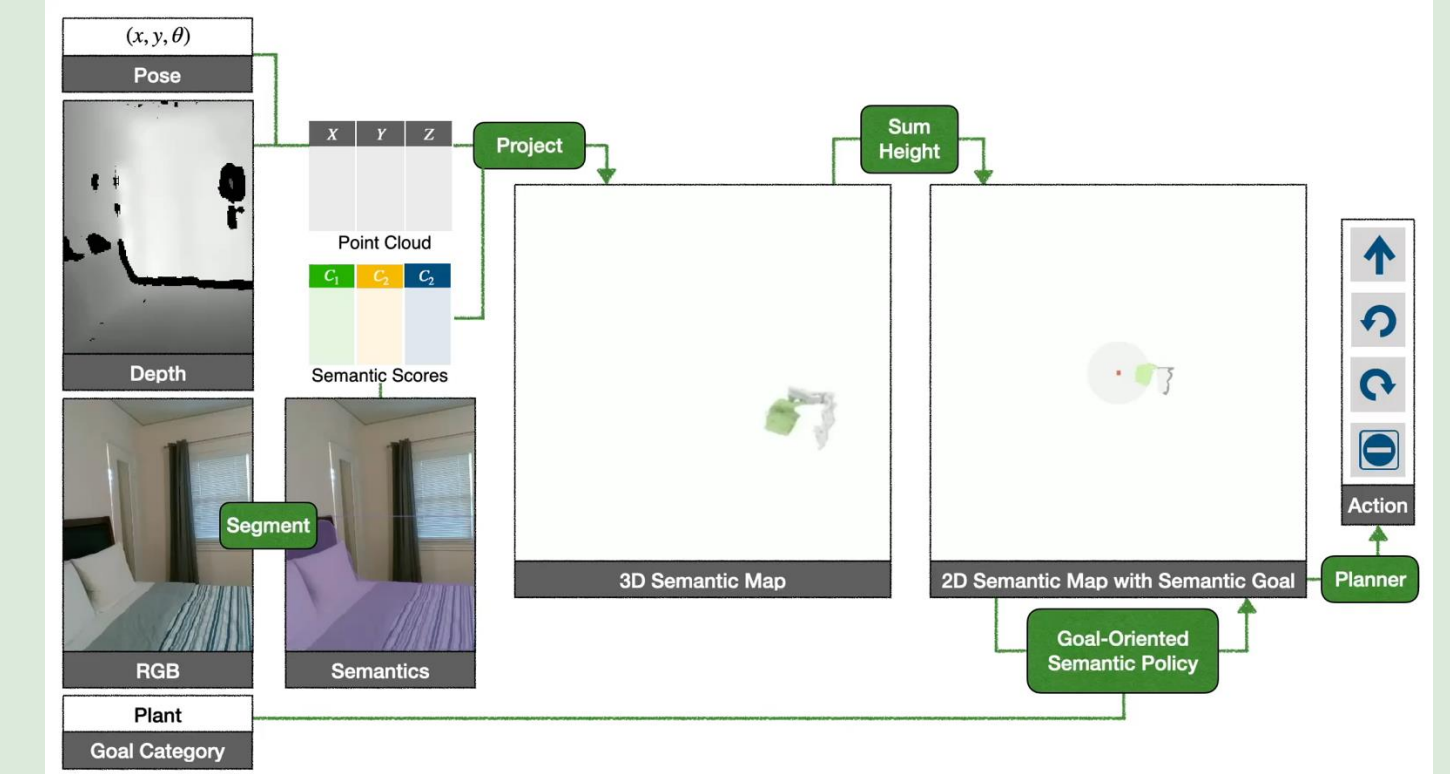
- Geometric Map
- Heuristic Exploration
 - No Training

End-to-end Learning



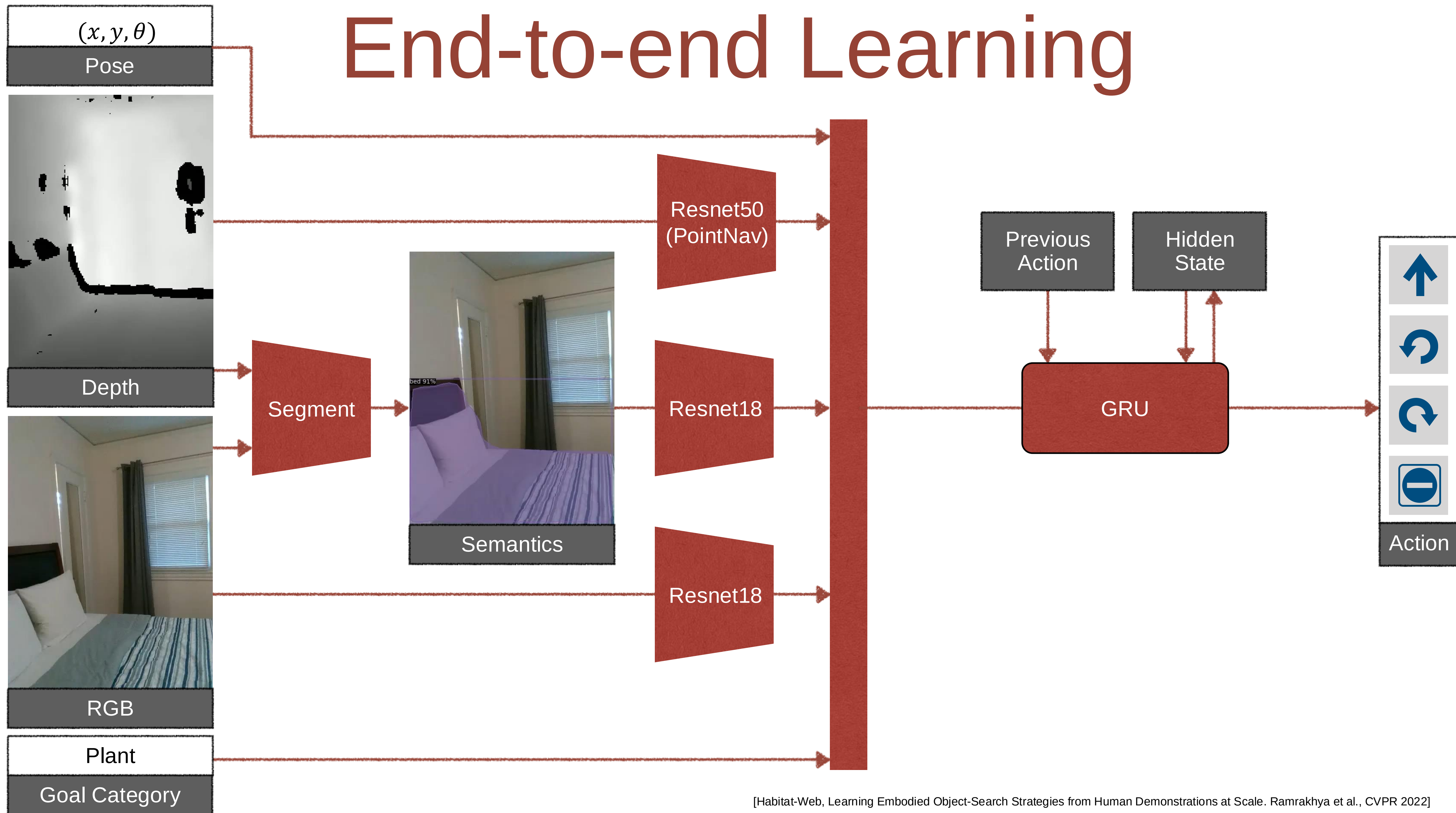
- End-to-end
- Large-scale IL + RL fine-tuning
 - 77,000 human trajectories
 - 200M frames of RL

Modular Learning

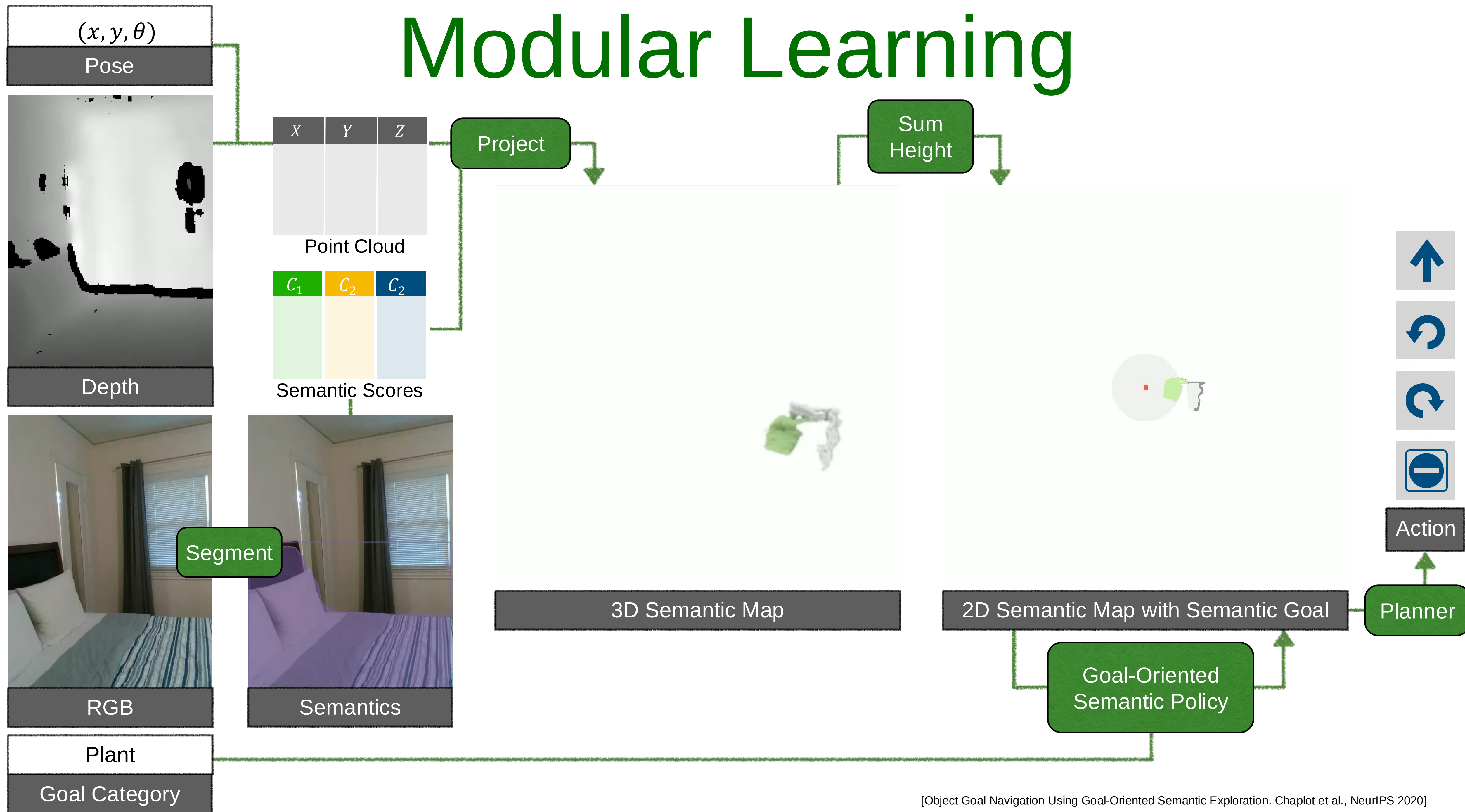


- Semantic Map
- Goal-Oriented Exploration
 - 10M frames of RL

End-to-end Learning



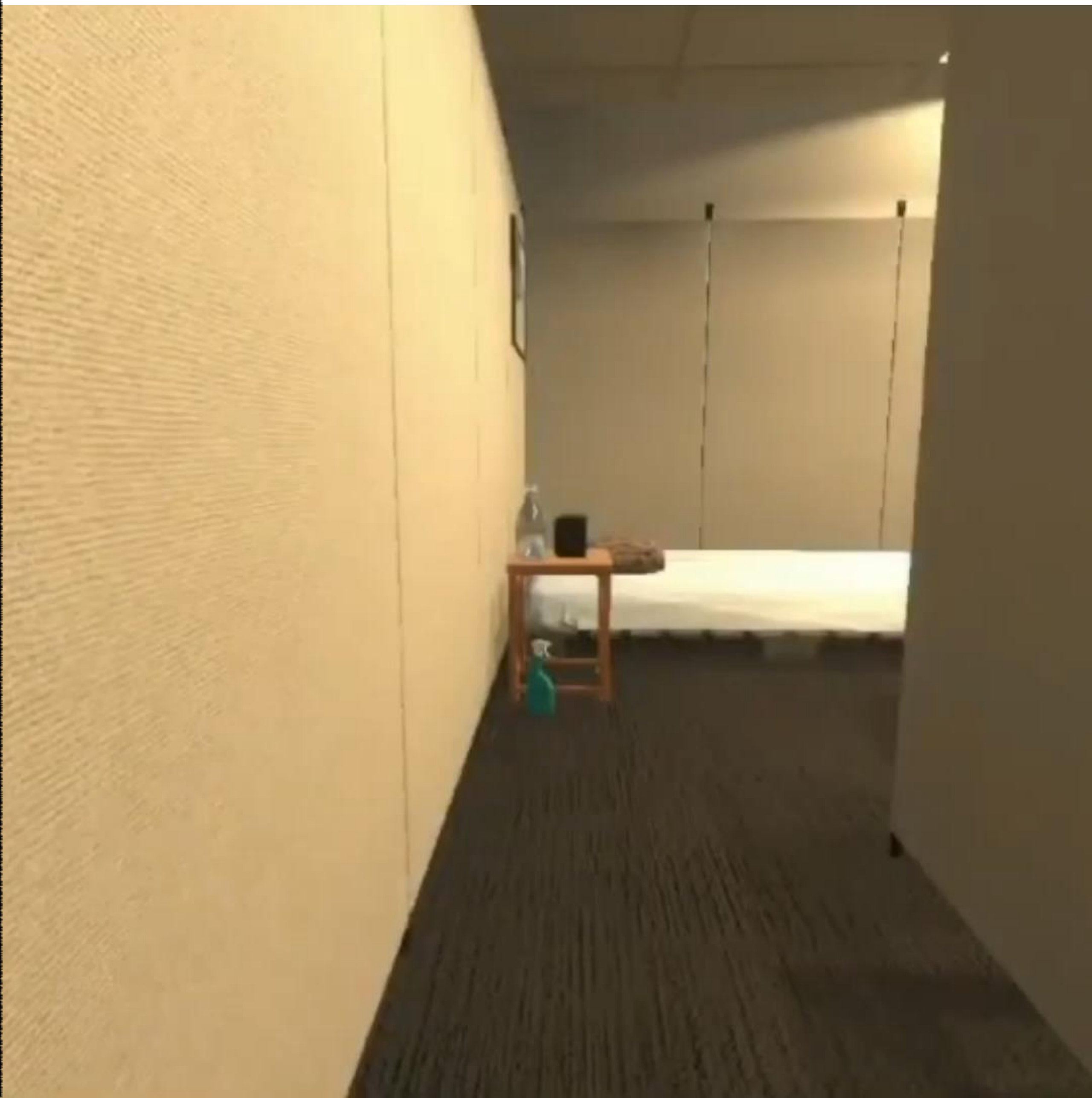
Modular Learning



Habitat



AI2-Thor



Results

Success Rate

SPL

■ Sim ■ Real World

Modular Learning

Classical

End-to-end Learning

0.00

0.25

0.50

0.75

1.00

Goal: *couch*

SPL: 0.74, 78 steps

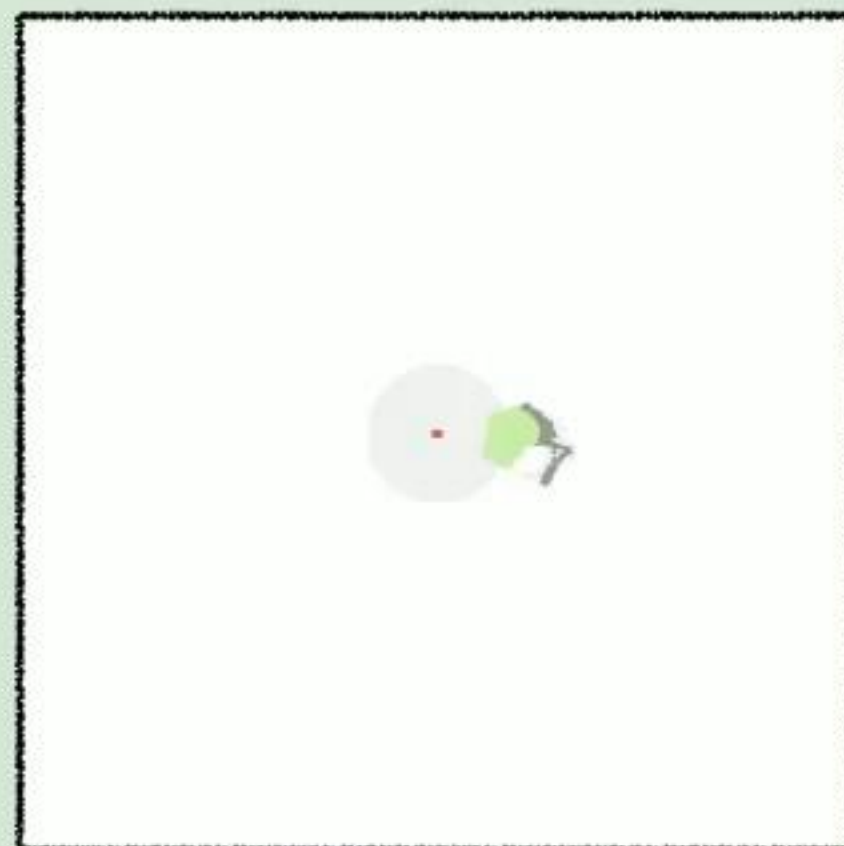
Modular

Third-person view

✓ Success



Observation



Predicted
Semantic Map

SPL: 0.0, 121 steps

End-to-End

Third-person view

✗ Failure



SPL: 0.33, 181 steps

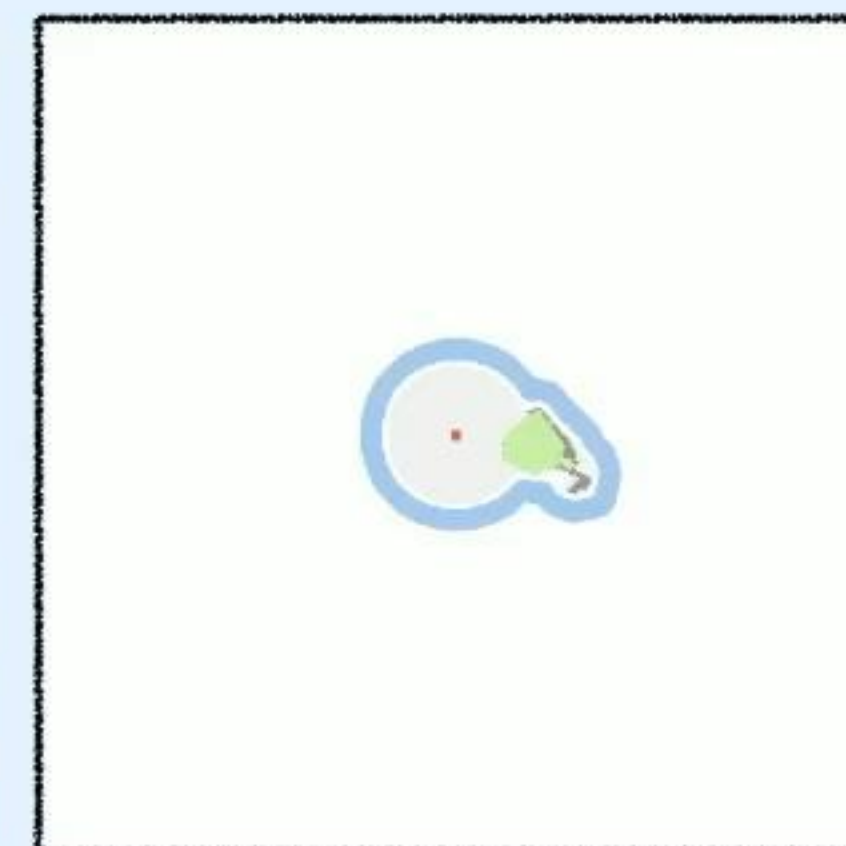
Classical

Third-person view

✓ Success

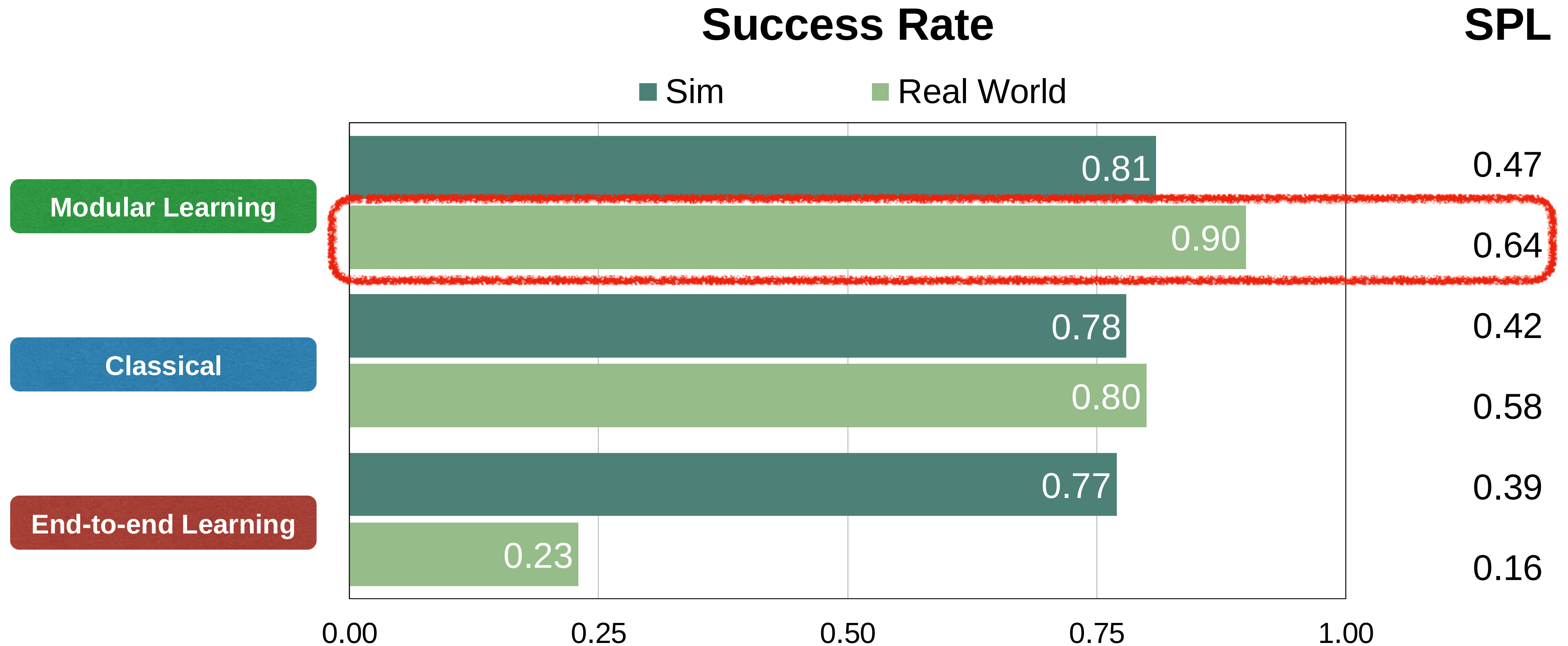


Observation



Predicted
Semantic Map

Modular Learning is Reliable



Observation



Goal: *toilet*

Predicted Semantic Map



Third-person view



— Navigable Area

— 0: chair

— 1: couch

— 2: potted plant

— 3: bed

— 4: toilet

— 5: tv

— 6: dining-table

— 7: oven

— 8: sink

— 9: refrigerator

— 10: book

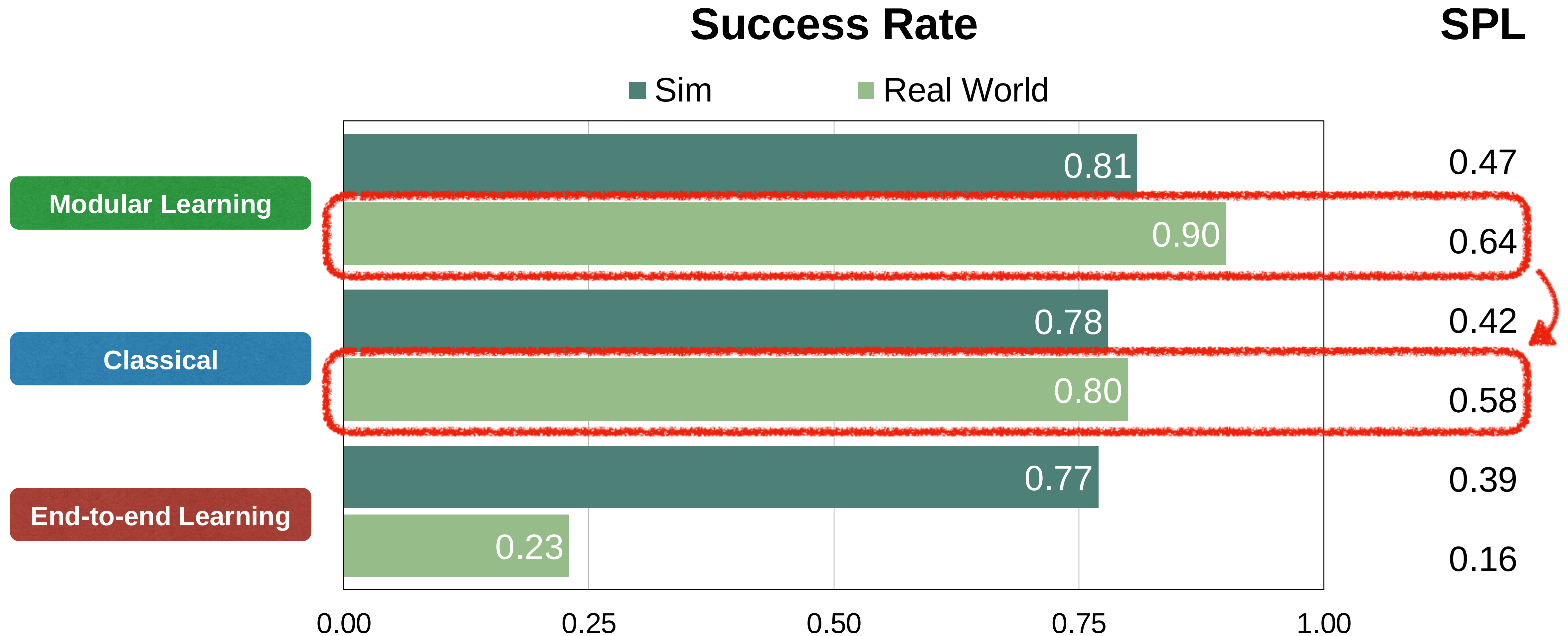
— 11: clock

— 12: vase

— 13: cup

— 14: bottle

Classical vs Modular Learning

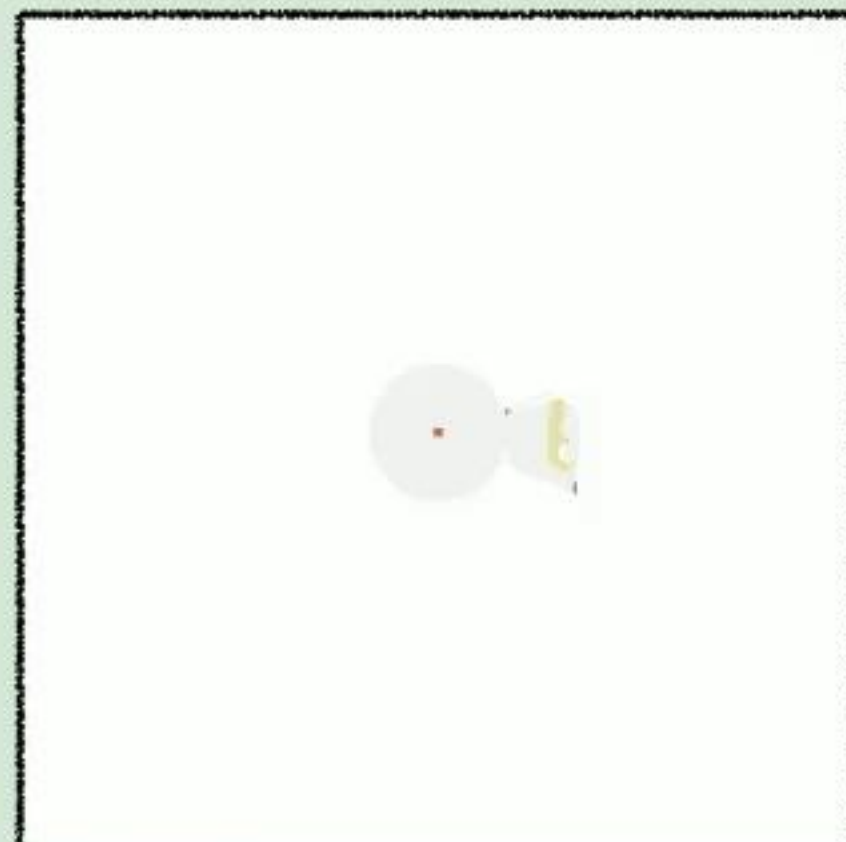


Classical vs Modular Learning

Goal: *bed*

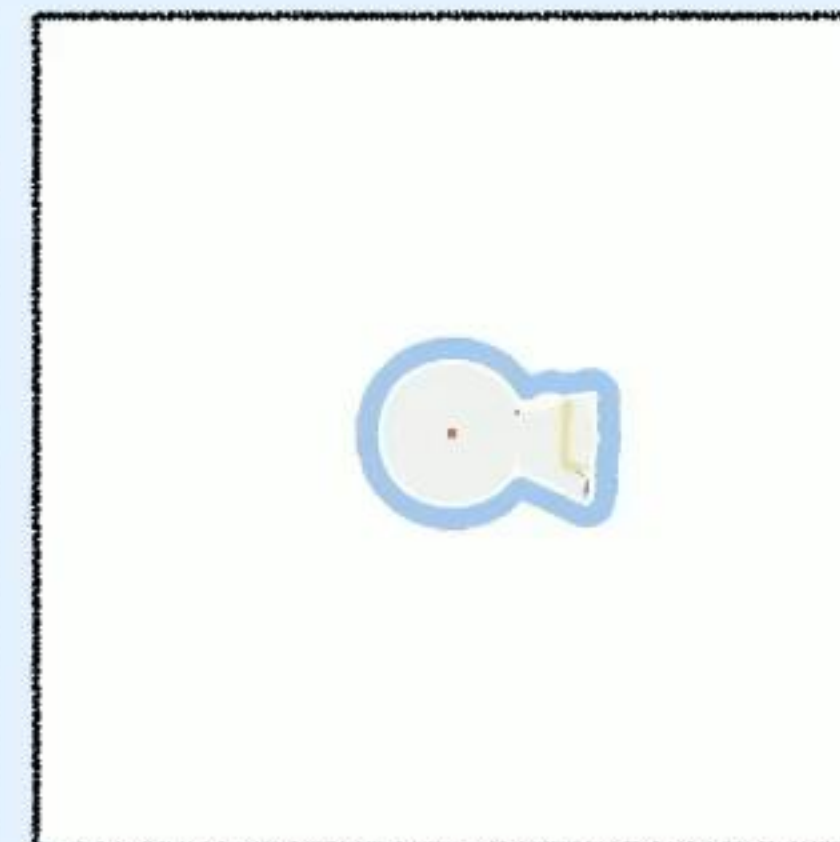
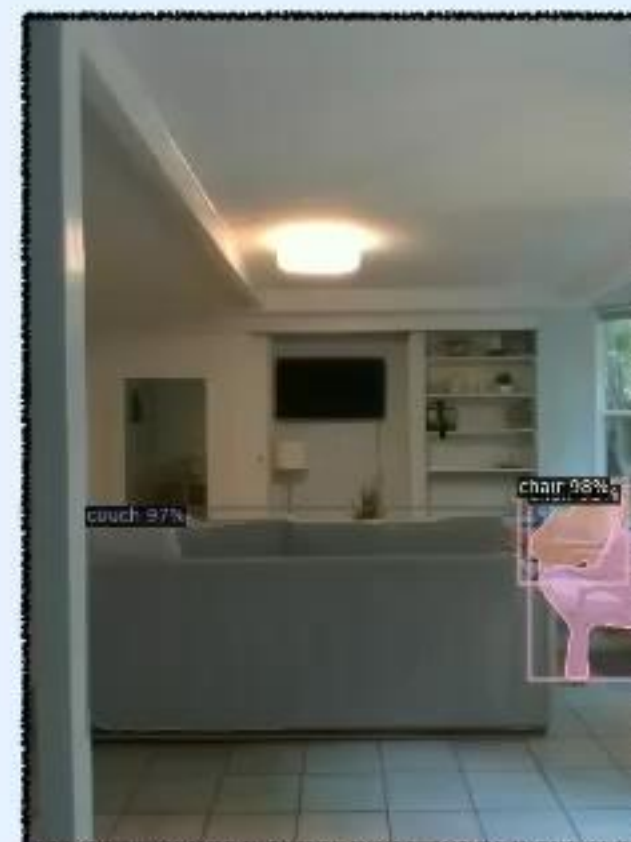
SPL: 0.90, 98 steps

Semantic Exploration



SPL: 0.52, 152 steps

Frontier Exploration



End-to-end fails to Transfer



End-to-end Failures

Goal: *TV*



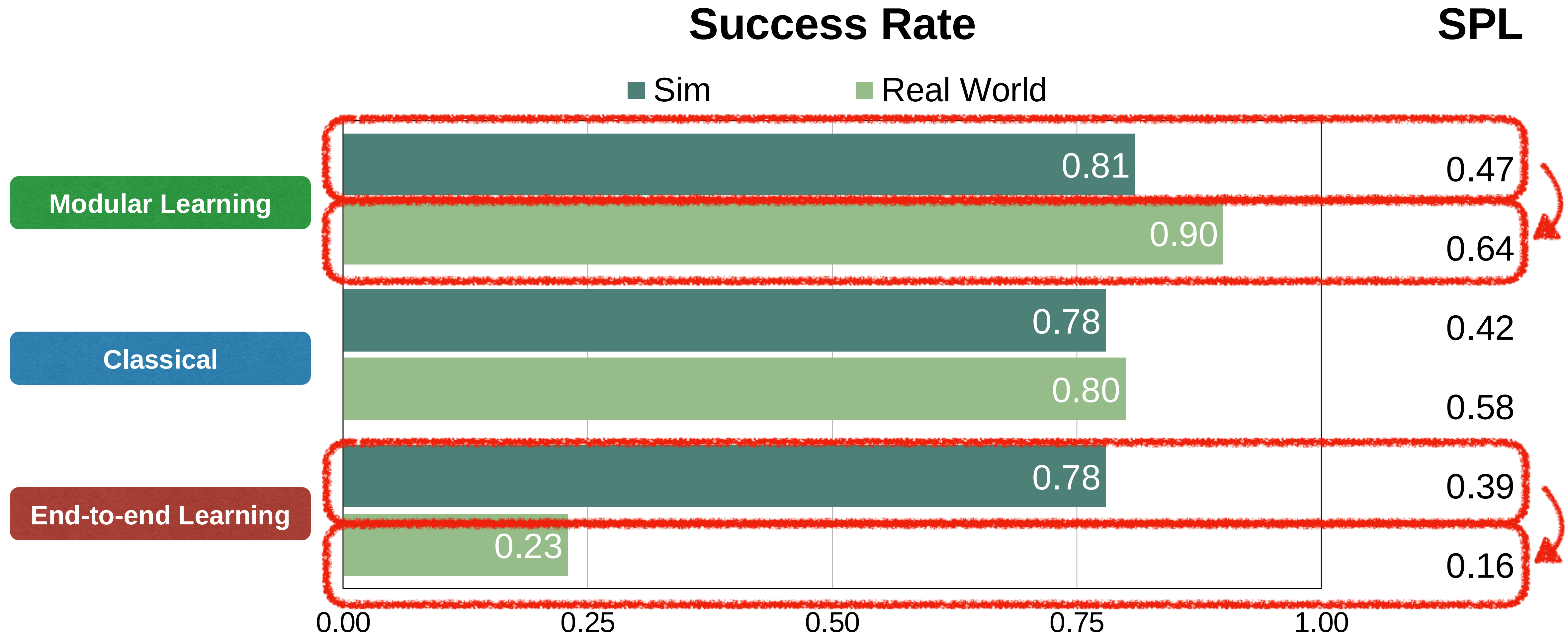
Goal: *Toilet*



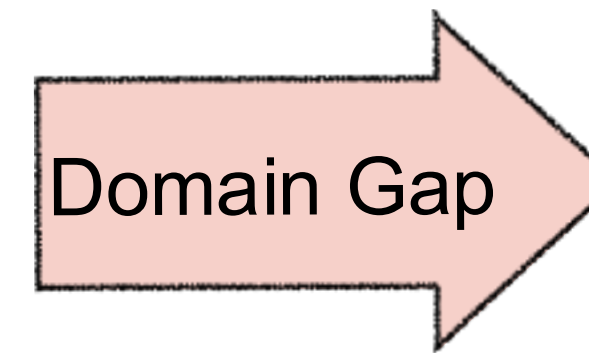
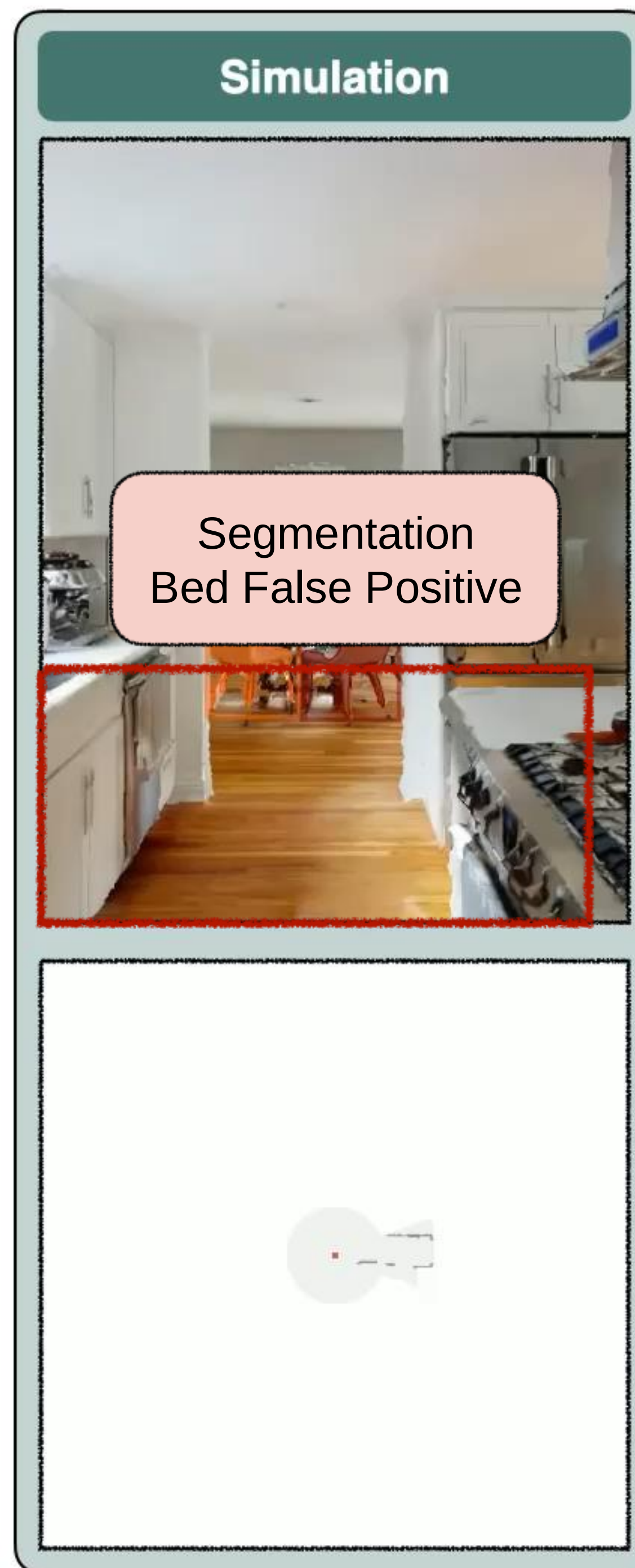
Goal: *Plant*



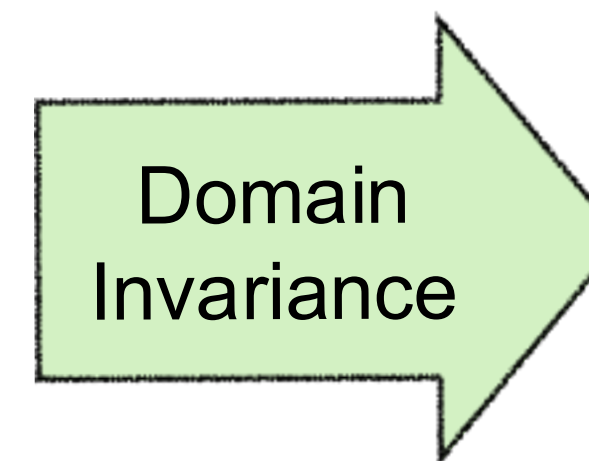
Modular vs End-to-end Transfer



Simulation vs Reality



*End-to-end
Learning
Input*

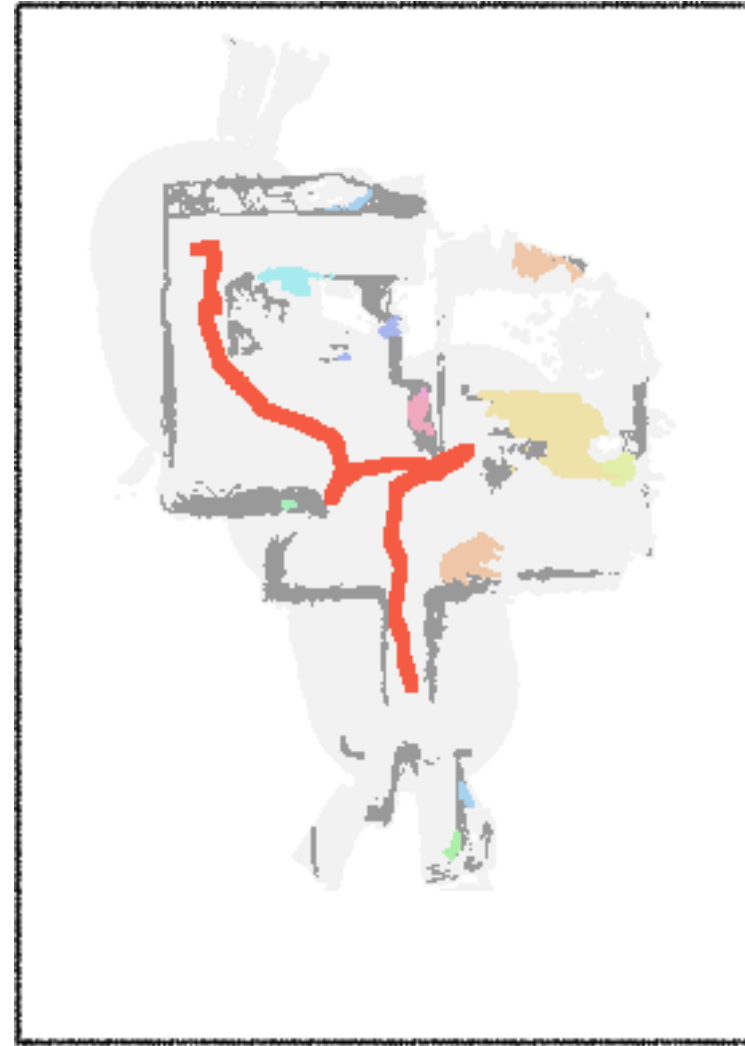


*Modular
Learning
Input*



Real World

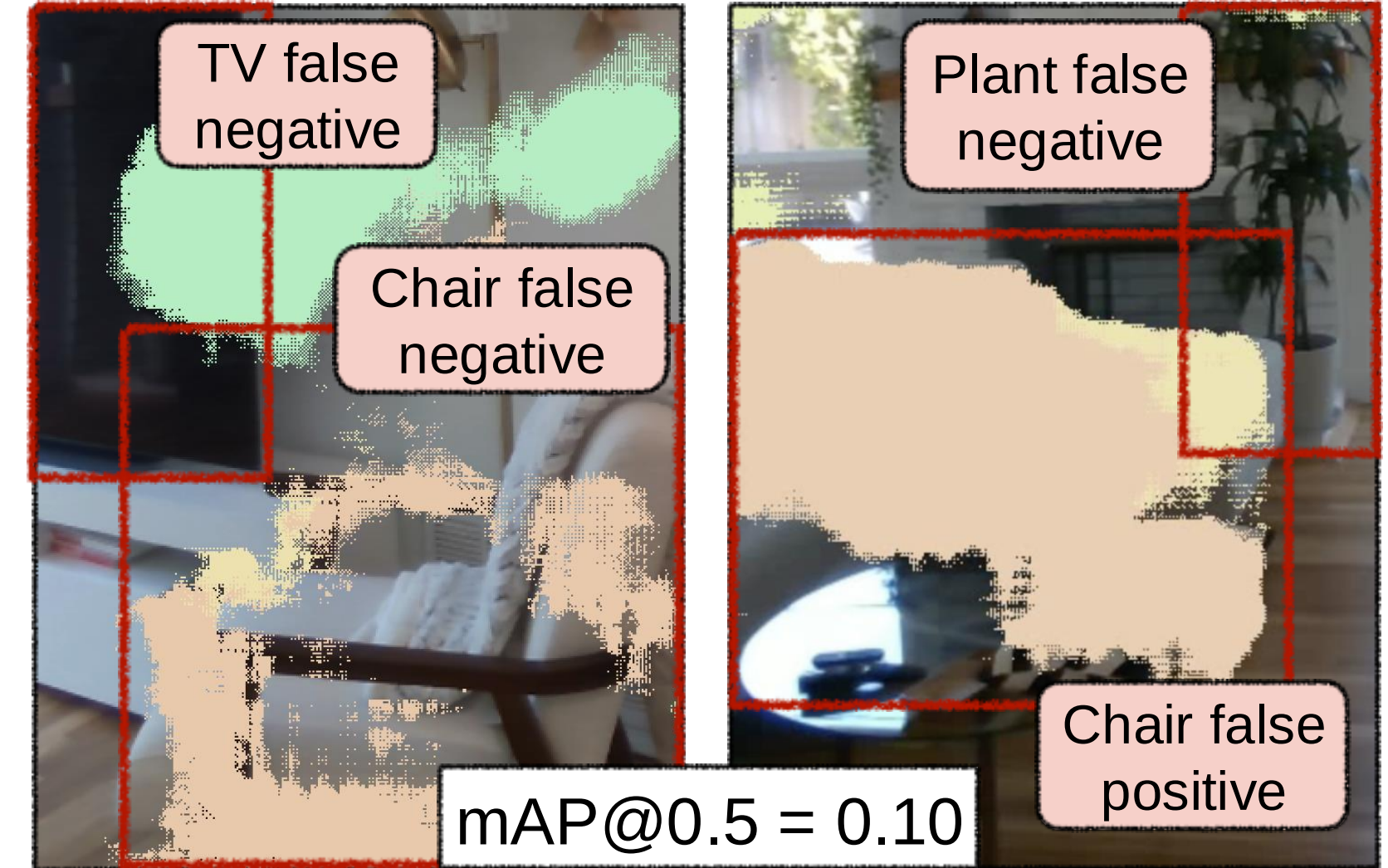
Predicted
Semantic Map



Segmentation Model
Trained in Real World



Segmentation Model
Trained in Simulation



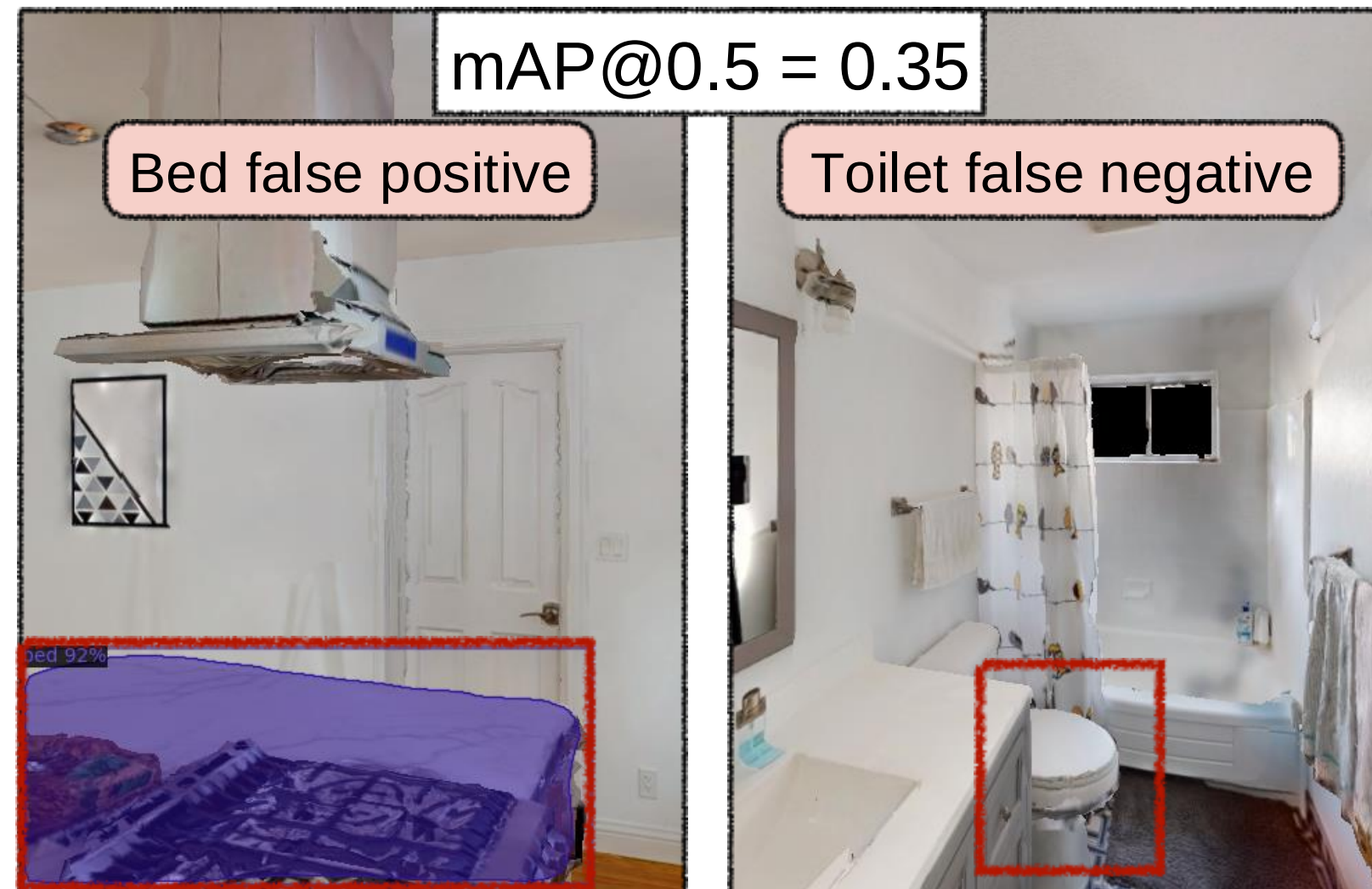
Domain
Invariance

Domain Gap

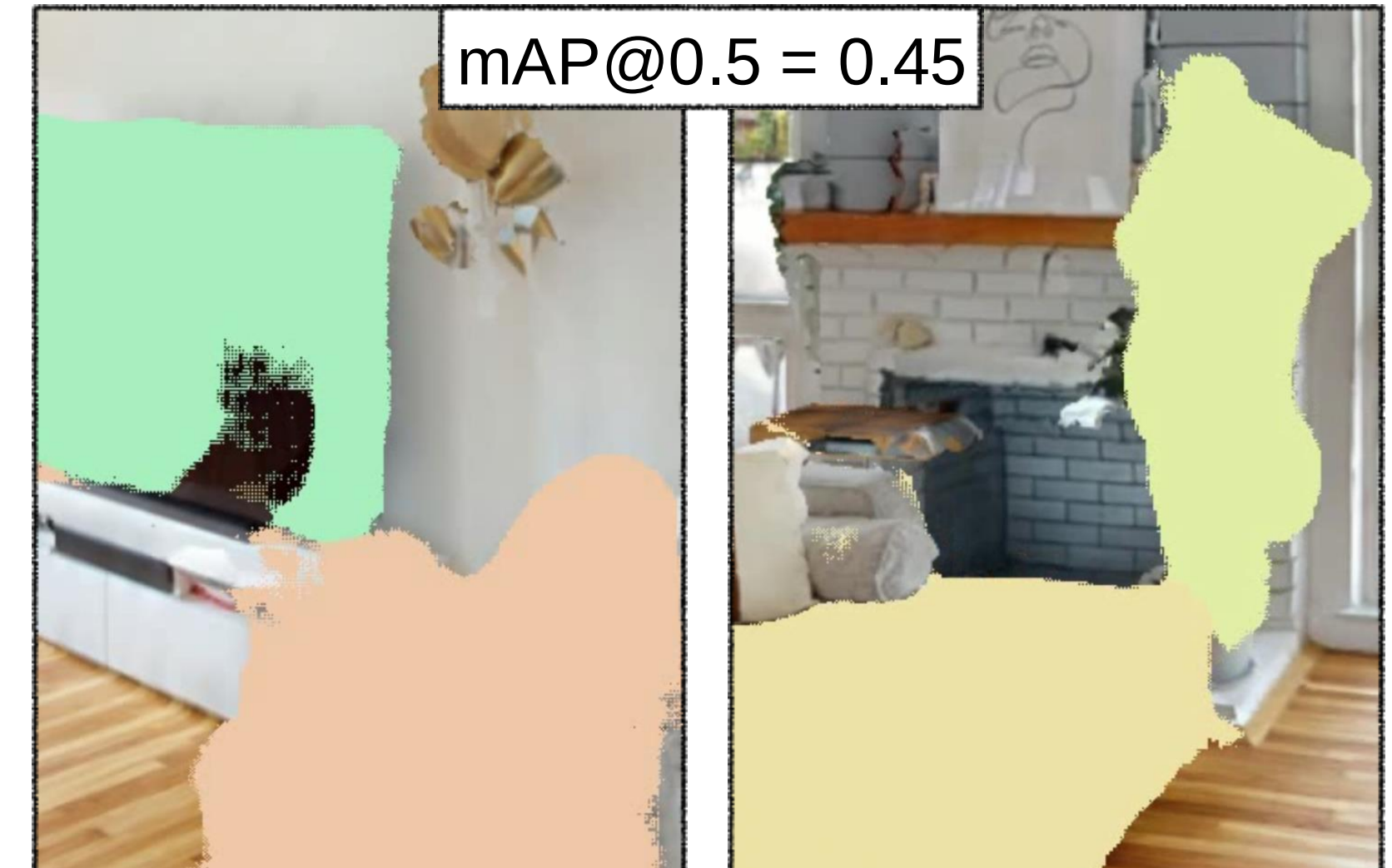
Domain Gap

- 0: chair
- 1: couch
- 2: potted plant
- 3: bed
- 4: toilet
- 5: tv

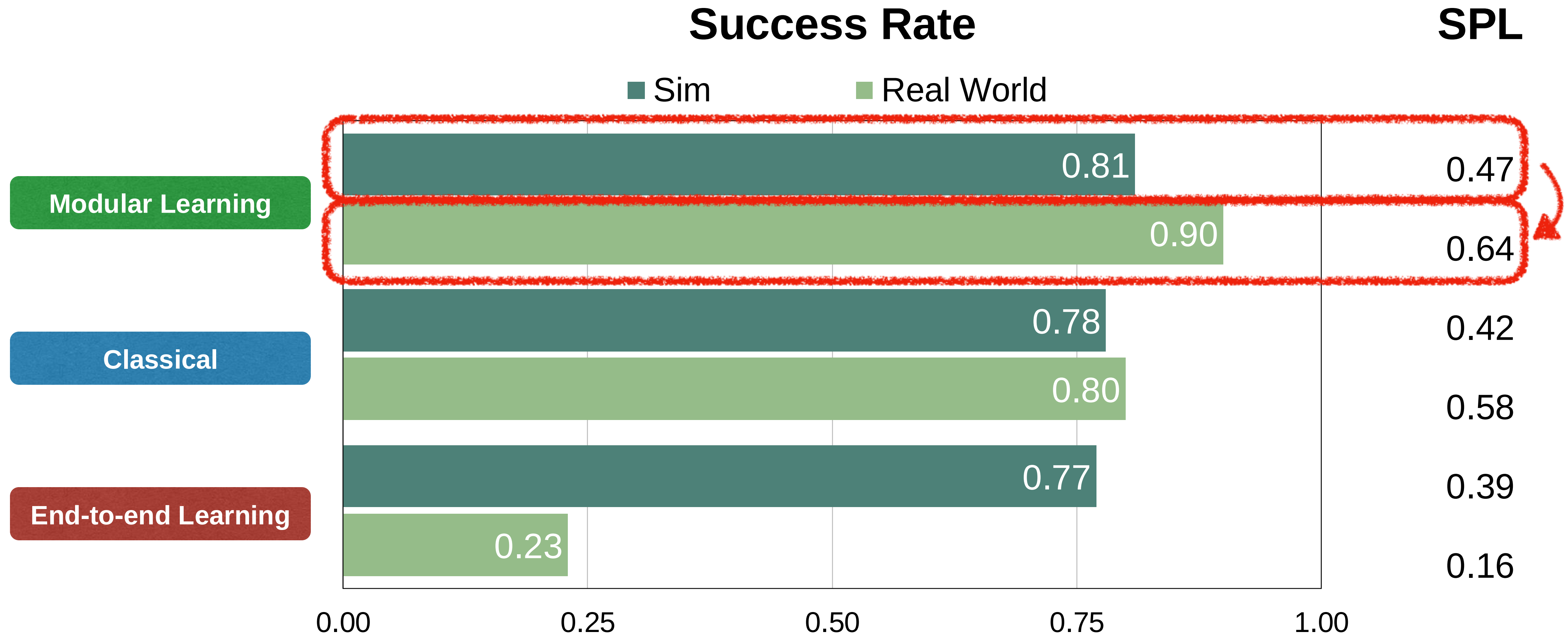
Simulation



mAP@0.5 = 0.45



Modular Learning Sim vs Real



Takeaways

For practitioners:

- Modular learning can reliably navigate to objects with 90% success

For researchers:

- Models relying on RGB images are hard to transfer from sim to real → *leverage modularity and abstraction in policies*
- Disconnect between sim and real error modes → *evaluate semantic navigation on real robots*